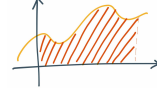


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# Mathematical Methods I

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Lectured in Autumn 2014 by Prof. A. J. MESTEL at Imperial College.  
Humbly typed by Karim BACCHUS.

Caveat Lector: unofficial notes. Comments and corrections should be sent to [kb514@ic.ac.uk](mailto:kb514@ic.ac.uk). Other notes available at [wwwf.imperial.ac.uk/~kb514](http://wwwf.imperial.ac.uk/~kb514).

## Syllabus

*The course supplies a firm grounding to A-level topics such as differentiation, integration, complex numbers and series expansions.*

*Functions:* Polynomial, rational, exponential, logarithmic, trigonometric and hyperbolic functions. Odd, even and inverse functions.

*Limits:* Basic properties and evaluation. Continuity & discontinuous functions.

*Differentiation:* First principles, differentiability; logarithmic and implicit differentiation; higher derivatives; Leibniz's formula; stationary points and points of inflexion; curve sketching; parametric representation, polar co-ordinates.

*Power Series:* The Mean Value Theorem. Taylor's Theorem with remainder. Infinite power series, radius of convergence. Ratio test; Taylor and Maclaurin Series. De l'Hopital's rule.

*Integration:* Definition as Riemann limit; indefinite & definite integrals; the fundamental theorem of calculus; integration by substitution and by parts; partial fractions; Existence of improper and infinite integrals. Integrals over areas and volumes.

*Complex Numbers:* Definition; the complex plane; standard and polar representation; de Moivre's Theorem;  $\exp(z)$  and  $\log(z)$

*First order Differentiation Equations:* Separable, homogeneous and linear equations. Special cases. Linear higher order equations with constant coefficients,

Course content at <http://www.ma.ic.ac.uk/~ajm8/M1M1>

## Appropriate books

G. Stephenson *Mathematical Methods for Science Students*

E. Kreyszig *Advanced Engineering Mathematics*

# Contents

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>0</b> | <b>Introduction</b>                                 | <b>3</b>  |
| <b>1</b> | <b>Functions</b>                                    | <b>4</b>  |
| 1.1      | Inverse functions . . . . .                         | 6         |
| 1.2      | Parity - Even & Odd Functions . . . . .             | 9         |
| 1.3      | Periodicity . . . . .                               | 11        |
| 1.4      | Polynomials . . . . .                               | 12        |
| 1.5      | Rational Functions . . . . .                        | 14        |
| <b>2</b> | <b>Infinite Series</b>                              | <b>18</b> |
| 2.1      | The Exponential Function . . . . .                  | 18        |
| 2.2      | Other Infinite Series . . . . .                     | 21        |
| 2.3      | Hyperbolic Functions . . . . .                      | 21        |
| 2.4      | Expanding other functions as power series . . . . . | 24        |
| 2.5      | Maclaurin Series . . . . .                          | 26        |
| 2.6      | Radius of Convergence . . . . .                     | 27        |
| <b>3</b> | <b>Differentiation</b>                              | <b>34</b> |
| 3.1      | First Principles Differentiation . . . . .          | 34        |
| 3.2      | Leibniz' Rule . . . . .                             | 40        |
| 3.3      | Four Theorems and an Example . . . . .              | 42        |
| 3.4      | De L'Hopital's Rule . . . . .                       | 44        |
| 3.5      | Taylor's Theorem . . . . .                          | 48        |
| 3.6      | Stationary Points . . . . .                         | 50        |
| 3.7      | Curve Plotting . . . . .                            | 52        |
| <b>4</b> | <b>Integration</b>                                  | <b>59</b> |
| 4.1      | The Riemann Integral . . . . .                      | 59        |
| 4.2      | Integrals over infinite ranges . . . . .            | 61        |
| 4.3      | Evaluation of Integrals . . . . .                   | 65        |
| 4.4      | Reduction Formulae . . . . .                        | 65        |
| 4.5      | Path Integrals . . . . .                            | 67        |
| 4.6      | Area of Integrals . . . . .                         | 70        |
| 4.7      | Volume Integrals . . . . .                          | 74        |
| <b>5</b> | <b>Complex Numbers</b>                              | <b>77</b> |
| 5.1      | Complex Logarithms . . . . .                        | 78        |
| 5.2      | Plotting Curves in the Complex Plane . . . . .      | 81        |
| <b>6</b> | <b>First order Differential Equations</b>           | <b>83</b> |

## 0 Introduction

**Example 0.1.** Consider the definition of the exponential function:

Lecture 0

$$\begin{aligned}\exp(x) &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \\ &= \sum_{n=0}^{\infty} \frac{x^n}{n!}\end{aligned}$$

So  $e^{100} = 1 + 100 + \frac{1}{2}100^2 + \frac{1}{6}100^3 + \dots \approx 0$

*Is this really true?*

Yes, but this is not obvious. The series converges (i.e. tends to an answer) for all  $x$ . We will see this later...

**Example 0.2.**  $f = \frac{4}{3 + \cos x}$

Can we write the power series  $f = a_0 + a_1x + a_2x^2 + \dots = \sum a_nx^n$  where  $a_n$  are known constants?

Yes, we find (somehow) that  $f = 1 + \frac{x^2}{8} + \frac{x^4}{192} + \dots$

Does this series converge? Using Maple to find the series and plotting  $f - \sum_0^{200} a_nx^n$ , we actually find that at  $\pm 3.60$ ish, the difference is non-zero and it fails to converge. This is (apparently) amazing evidence of the existence of the complex plane...

**Example 0.3** (Limits).

$\lim_{x \rightarrow \infty} (\sin x) = ?$ , it is undefined.

$\lim_{x \rightarrow \infty} \left( \frac{\sin x}{x} \right) = 0$ , from the sandwich theorem as 0 is squeezed between  $\frac{1}{x}$  and  $-\frac{1}{x}$

$\lim_{x \rightarrow \infty} \left( \frac{1}{x \sin x} \right) = ?$ , undefined once more, since whenever  $x = n\pi$ , the denominator is 0.

$\lim_{n \rightarrow \infty} \left( \frac{\sin n}{n} \right)$ ,  $n \in \mathbb{R}$  is not at all obvious (depends on how well you can approximate  $\pi$ )

Clearly we have work to do....

# 1 Functions

A function takes an “input” and gives a *unique* output:  $f(x) : x \in \mathbb{R}$ .  $f(x)$  is the output or function value at  $x$ .  $f : \mathbb{R} \rightarrow \mathbb{R}$  (alternative notation:  $f$  “maps” input  $\in \mathbb{R}$  to output  $\in \mathbb{R}$ ).  $f$  may not be defined for all reals. A function *should* be defined along with the *domain* of values over which it applies, e.g.:

Lecture 1

$$f(x) = \sqrt{x^2 - 1} \text{ for } x \geq 1$$

**Definition.**  $[a, b]$  means  $\forall x : a \leq x \leq b$ ,  $(a, b)$  means  $\forall x : a < x < b$ . These are called *closed* and *open* intervals respectively.

So  $x \geq 1$  could be written as  $x \in [1, \infty)$ . By convention  $\infty$  is never a closed interval since it is not a real number.

**Definition.** We also define the *range* of a function to be the set of possible values  $f(x)$  as it takes values of the domain.

So  $f(x) = \sqrt{x^2 - 1}$  in  $[1, \infty)$  has the *range*  $[0, \infty)$ .

Note:  $\sqrt{\phantom{x}}$  is always positive conventionally, otherwise it maps to more than one value  $\implies f$  is not a function. Hence  $\sqrt{x^2} = |x|$ , *not necessarily*  $x$ .

*How might we define functions?*

- (i) An explicit formula, e.g.  $f(x) = x^2 \sin(x)$

As the domain is not given, we assume it applies for all  $x$  or all sensible  $x$ .

e.g.  $f(x) = \frac{x+2}{x-1}$  “sensible” here means  $x \neq 1$

- (ii) Split ranges, e.g.

$$f(x) = \begin{cases} x & \text{if } x > 1 \\ \sin(x^2) & \text{if } 0 < x \leq 1 \\ e^x & \text{if } x \leq 0 \end{cases}$$

- (iii) As a solution to an equation

e.g.  $f'' + x^2 f = 0$ ,  $f(0) = 1$ ,  $f'(0) = 0$  *may* define a function

Similarly we could define  $f(x) = \int_0^x t^t dt$

(Note: we use a different letter for the *dummy* variable,  $t$ )

- (iv) In words, e.g.  $f(x) =$  “the maximum amount by which  $x$  exceeds an integer for  $n = 1, 2, \dots, 100$ ”

- (v) An implicit definition, e.g.  $f(x)$  given by  $f(x) + \frac{1}{2} \sin[f(x)] = x$   
 (or  $y + \sin y = x$  : we can't solve for  $y$  in terms of  $x$  easily) given  $x$ , not easy to calculate  $f(x)$

- (vi) As a limit, e.g.  $x^{x^{x^{\dots}}}$  or more formally:  $f_1 = x^x$ ,  $f_{n+1} = x^{f_n}$  for  $n \geq 1$

If this process tends to a limit as  $n \rightarrow \infty$  we may have defined a function.

... And so on. There are lots of ways of defining functions.

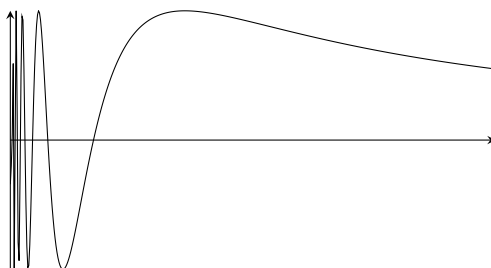
*How many functions are there?*

It turns out it's (a very large...) infinity (see M1F, there are different sizes of "infinities")

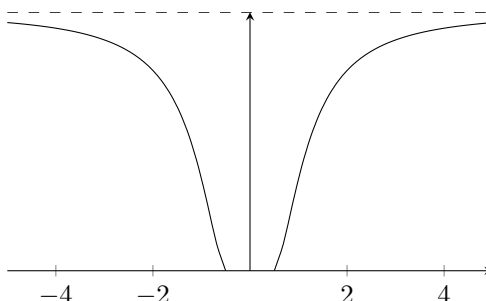
Most functions are ~~horrible~~ horrendous. Even ones which look nice can be nasty...

**Example 1.1.**  $f(x) = \begin{cases} \sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$

The graph crosses the  $x$ -axis an infinite number of times between  $[0, n]$ :



**Example 1.2.**  $f(x) = e^{-\frac{1}{x^2}}$



$f(x)$  is so flat at zero that the Maclaurin (Taylor) Series converges to 0. This is *not* the right answer. We then call this a non-analytical function.

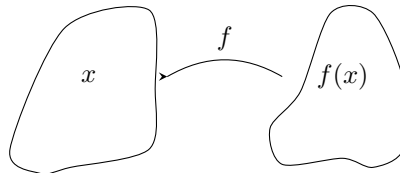
**Example 1.3.**  $f(x) = \sum_{n=1}^{\infty} \frac{\sin(n^4 x)}{n^2} = \sin x + \frac{\sin 16x}{4} + \frac{\sin 81x}{9} + \dots$

This function is continuous everywhere, differentiable nowhere.

## 1.1 Inverse functions

Suppose we have a function  $f$

Lecture 2



**Definition.** If it is possible to find a function  $g(f(x)) = x \quad \forall x$  in domain of  $f$ , then  $g$  is called the *inverse* of  $f$ . It's often denoted as  $f^{-1}$ .

The domain of  $f =$  the range of  $f^{-1}$ , and the range of  $f =$  the domain of  $f^{-1}$

*Do inverses always exist?*

Clearly not if  $(\geq)$  two  $x$  values give the same value of  $f(x) = y$  say. As we cannot determine a unique  $x$  value given  $y$ . In practice, we try to solve  $f(x) = y$  for  $x$ . This may find the inverse or tell us that there is a problem.

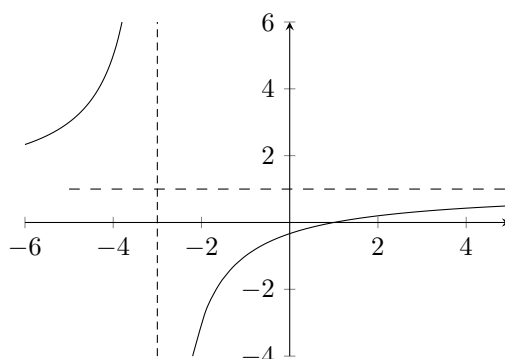
**Example 1.4.** Find the inverse of  $f(x) = \frac{x-1}{x+3} \quad (x \neq -3)$

$$y = \frac{x-1}{x+3}$$

$$\implies y(x+3) = x-1$$

$$\implies x(y-1) = -3y-1$$

$$\implies x = \frac{3y+1}{1-y} = f^{-1}(y)$$



The graph  $y = f(x)$  helps us understand what is going on. So we sketch it, noting that  $y = \frac{x-1}{x+3} = 1 - \frac{4}{x+3}$ . For inverse to exist, all lines  $y = \text{constant}$  must intersect  $y = f(x)$  ~~once~~ once and only once, which is clearly the case.

**Example 1.5.** Does  $f^{-1}$  exist for  $y = x + \frac{1}{x} = f(x)$ ?

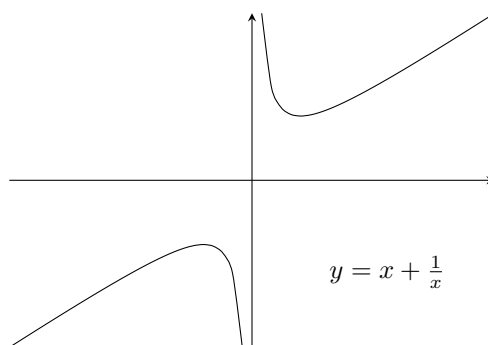
Note that if  $x = 2$ , then  $f(2) = 2.5$ , but  $f(\frac{1}{2}) = 2.5$  also... maybe if we restrict the domain of  $f$ , we can find a sensible inverse. Lets try  $y = x + \frac{1}{x}$

$$\begin{aligned} \implies xy &= x^2 + 1 \\ \implies x^2 - xy + 1 &= 0 \\ \implies x &= \frac{y \pm \sqrt{y^2 - 4}}{2} \end{aligned}$$

Which root should we take?

Sketching  $f(x)$ , we can note that  $y = k$  intersects the graph:

$$\begin{cases} \text{Not at all} & -2 < k < 2 \\ \text{Once} & k = \pm 2 \\ \text{Twice} & k > 2 \text{ or } k < -2 \end{cases}$$



Suppose we restrict the domain of  $f$  to be  $|x| \leq 1$  (excluding  $x = 0$ ).  
Then we can define the inverse function:

$$f^{-1}(y) = \begin{cases} \frac{y - \sqrt{y^2 - 4}}{2} & \text{if } y \geq 2 \\ \frac{y + \sqrt{y^2 - 4}}{2} & \text{if } y \leq 2 \end{cases}$$

If instead we restrict the domain of  $f$  to be  $|x| \geq 1$  then:

$$f^{-1}(y) = \begin{cases} \frac{y + \sqrt{y^2 - 4}}{2} & \text{if } y \geq 2 \\ \frac{y - \sqrt{y^2 - 4}}{2} & \text{if } y \leq 2 \end{cases}$$

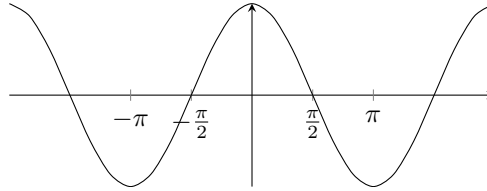
## Trigonometric Functions

*Trigonometry:* trigōnon metron - The Measuring of Triangles

Later we will define  $\cos(x)$ ,  $\sin(x)$ ,  $\tan(x)$ , but you already know them.

### Inverse Cosine

If  $f(x) = \cos(x)$  then  $f^{-1}(x) = \cos^{-1}(x)$  (or  $\arccos(x)$ ) exists for some  $x$  and some agreed domain of  $f(x)$ .

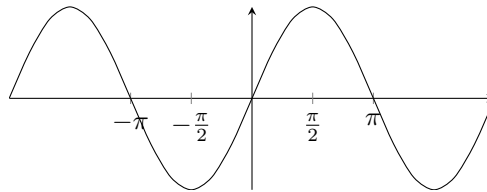


The natural domain to restrict is  $0 \leq x \leq \pi$ . Then  $\cos^{-1}(y)$  exists uniquely, provided  $|y| \leq 1$ . Now remove the restrictions on  $x$ . Solve the equation  $\cos x = \alpha$ :

General solution:  $x = 2n\pi \pm \cos^{-1} \alpha \quad (n \in \mathbb{Z})$

But remember  $0 \leq \cos^{-1} \leq \pi$  always!

### Inverse Sine



We restrict  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$  i.e.  $|x| \leq \frac{\pi}{2}$ . Then the inverse function  $\sin^{-1} y$  exists, if  $|y| \leq 1$ . So  $-\frac{\pi}{2} \leq \sin^{-1} \leq \frac{\pi}{2}$ . Then the equation  $\sin x = \beta$  has the general

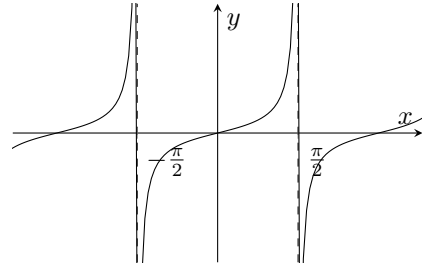
solution:

$$\begin{aligned} x &= 2n\pi + \sin^{-1} \beta \text{ if } n \text{ is even} \\ x &= 2n\pi - \sin^{-1} \beta \text{ if } n \text{ is odd} \\ \implies \boxed{x &= 2n\pi + (-1)^n \sin^{-1} \beta \quad (n \in \mathbb{Z})} \end{aligned}$$

### Inverse Tan

We restrict  $-\frac{\pi}{2} \leq \tan^{-1} y \leq \frac{\pi}{2}$  defined  $\forall y$ .  
So finally  $\tan x = \gamma$  has general solution:

$$\boxed{x = n\pi + \tan^{-1} \gamma}$$



## 1.2 Parity - Even & Odd Functions

**Definition.** A function  $f(x)$  defined over a symmetric domain (i.e.  $[-a, a]$ ) is called even  $\iff f(-x) = f(x)$  and odd  $\iff f(-x) = -f(x)$

Lecture 3

e.g.  $x^2$  is even,  $\sin x$  is odd. Functions need not be even or odd. But any function (over a symmetric domain) can be written as the sum of an even function & an odd function.

#### Example 1.6.

$$\frac{x}{x+1} = \frac{x}{x+1} \frac{(x-1)}{(x-1)} = \underbrace{\frac{x^2}{x^2-1}}_{\text{Even}} - \underbrace{\frac{x}{x^2-1}}_{\text{Odd}}$$

#### Example 1.7.

$$\cos(x+3) = \underbrace{\cos x \cos 3}_{\text{Even}} - \underbrace{\sin x \sin 3}_{\text{Odd}}$$

In general, how do we write  $f(x) = f_e(x) + f_o(x)$ ?

$$f(x) = f_e(x) + f_o(x) \tag{1}$$

$$\implies f(-x) = f_e(-x) + f_o(-x)$$

$$\implies f(-x) = f_e(x) - f_o(x) \tag{2}$$

Solving (1) and (2) by adding:

$$\implies f_e(x) = \frac{1}{2}[f(x) + f(-x)]$$

Similarly

$$f_o(x) = \frac{1}{2}[f(x) - f(-x)]$$

To prove that we can always find these two functions, start again from other way:

Define  $f_e$  and  $f_o$  as above, note:

- (i)  $f_e$  is even
- (ii)  $f_o$  is odd
- (iii)  $f_e + f_o = f$

This proves that any  $f$  has an even part and an odd part.

**Example 1.8.** Redo Example 1.6.  $f(x) = \frac{x}{x+1} = f_e(x) + f_o(x)$

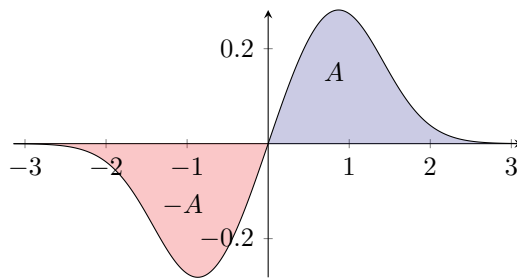
$$\Rightarrow f_e(x) = \frac{1}{2} \left[ \frac{x}{x+1} + \frac{-x}{1-x} \right] = \frac{1}{2} \left[ \frac{x - x^2 - x^2 - x}{(1+x)(1-x)} \right] = \frac{x^2}{x^2 - 1}$$

## Evaluating Integrals

Parity is a great help when evaluating integrals.

What is  $\int_{-\pi}^{\pi} \frac{x + \sin(x^3)}{1 + e^{x^2}} dx$ ?

Replacing  $x$  by  $-x$  we can see that the integrand is an odd function since  $f(-x) = -f(x)$ . Hence the Integral = 0.



**Proposition 1.9.** In general  $I = \boxed{\int_{-a}^a f_o(x) dx = 0}$

*Proof.* Substituting  $t = -x$

$$\begin{aligned}
\Rightarrow I &= \int_a^{-a} f_o(-t) (-dx) \\
&= \int_{-a}^a f_o(-t) (dx) \quad [\text{Use - sign to swap limits}] \\
&= -I
\end{aligned}$$

Hence  $I = -I \Rightarrow I = 0$ .

We conclude that if  $f(x)$  is odd, then  $\int_{-a}^a f(x) dx = 0$ . ■

Later we will deal with power series  $f(x) = a_0 + a_1x + a_2x^2 + \dots$ , where  $a_i$  is a given constant for  $i \in \mathbb{N}$ .

If  $f(x)$  is even then  $a_1 = 0, a_3 = 0$  etc., i.e.  $a_{\text{odd}} = 0$ . If  $f(x)$  is odd then  $a_0 = 0, a_2 = 0$  etc., i.e.  $a_{\text{even}} = 0$ . So even/odd functions only have even/odd powers of  $x$ .

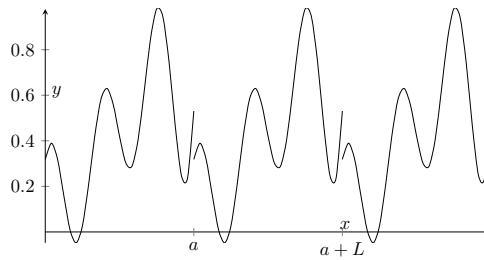
### 1.3 Periodicity

**Definition.** We say a function  $f(x)$  is  $T$ -periodic if and only if  $f(x + T) = f(x) \forall x$ , where  $T > 0$  and  $T$  is the smallest value for which this holds.

So although  $\sin(x + 4\pi) = \sin x \forall x$ , we do not say that  $\sin x$  is  $4\pi$  periodic, as  $\sin(x + 2\pi) = \sin(x)$  as well.

| $f(x)$           | period                                                    |
|------------------|-----------------------------------------------------------|
| $\cos^2 x$       | $\pi$                                                     |
| $\cos  x $       | $2\pi$                                                    |
| $ \cos x $       | $\pi$                                                     |
| $\sin(\alpha x)$ | $\frac{2\pi}{\alpha} \quad (\alpha \neq 0)$               |
| 3                | depends on definition, say period = 0 & change definition |
| $\sin  x $       | not periodic                                              |
| $ \sin x $       | $\pi$                                                     |

*Are there any other periodic functions (other than the trigonometric ones)?*



We can turn any function into a periodic one. Any function defined on a finite interval can be extended into a periodic function over all  $\mathbb{R}$  by copying. Define  $f(x+L) = f(x)$  to replicate the behaviour  $\forall x$ .

## 1.4 Polynomials

**Definition.** An  $n$ th order polynomial in  $x$  is a function of the form:

Lecture 4

$$f(x) = \sum_{n=0}^N a_n x^n$$

where  $a_N \neq 0$ .  $N$  is called the *degree* or *order* of the polynomial.  $a_n$  for  $n = 0, \dots, N$  are called the coefficients. If  $a_n$  is real  $\forall n$ , we say the polynomial is real (even if  $x$  may be complex).

### Theorem 1.10: The Fundamental Theorem of Algebra

Every polynomial has a root (possibly complex). In general, we call a value  $\alpha$  a root of  $f(x)$  if  $f(\alpha) = 0$ .

*Proof.* See next year's course M2PM3 COMPLEX ANALYSIS.

**Corollary 1.11.** If  $c$  is a root of an  $N$ th order polynomial  $P_N(x)$ , then we can write  $P_N(x) = (x - c)P_{N-1}(x)$ .

**Corollary 1.12.** Every  $N$ th order polynomial has precisely  $N$  roots, allowing for repeated roots. e.g.  $(x - 1)^2$  has roots  $1, 1$ .

**Corollary 1.13.** If  $P(x)$  is a real polynomial with a complex root  $\alpha + i\beta$  ( $\alpha, \beta$  real,  $\beta \neq 0$ ), then it also has root  $\alpha - i\beta$  (the complex conjugate).

**Corollary 1.14.** Every real polynomial can be written as:

$$P_N(x) = A(x - r_1)(x - r_2)\dots(x - r_M)((x - \alpha_1)^2 + \beta_1^2)((x - \alpha_2)^2 + \beta_2^2)\dots((x - \alpha_L)^2 + \beta_L^2)$$

Where  $r_1..r_M$  are the real roots, and  $(\alpha \pm i\beta), \dots (\alpha_L \pm i\beta_L)$  are the complex roots and  $M + 2L = N$

i.e. Any real polynomial can be written as a product of real linear and quadratic factors.

N.B. If the polynomial is not real (i.e. if at least one coefficient is strictly complex), then the complex roots need not be in conjugate pairs.

## Roots of polynomials

- Linear:  $ax + b = 0$ , one trivial root
- Quadratic:  $ax^2 + bx + c = 0$

$$\text{Formula: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{OR} \quad x = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}}$$

*Exercise 1:* Show these are the same.

*Exercise 2:* Use your calculator to solve the equation  $\epsilon x^2 + (1 + \epsilon)x + 1 = 0$ , where  $\epsilon$  is very small (i.e.  $10^{-12}$ ).

Subtracting two numbers which are very close together leads to severe accuracy loss c.f. Patriot missiles. The “Best” formula to solving a quadratic depends on  $a, b$  and  $c$  in practice.

- Cubics:  $ax^3 + bx^2 + cx + d = 0$

There is a formula (see M1F), but it has very little practical use.

- Quartics:  $ax^4 + bx^3 + cx^2 + dx + e = 0$

There is also a general formula... once again not worth knowing.

- Quintics:  $ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$

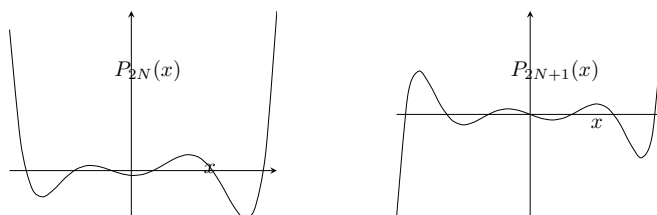
There is no general formula to express the roots in terms of radicals. See M3P11 GALOIS THEORY in 3rd year. But one can easily find the roots in practice for any particular case.

- $N > 5$ : Similarly no formula. However considering general  $N$ :

$$\begin{aligned} P_N(x) &= a_N x^N + a_{N-1} x^{N-1} + \dots \\ &= a_N x^N \left[ 1 + \frac{a_{N-1}}{a_N} \frac{1}{x} + \frac{a_{N-2}}{a_N} \frac{1}{x^2} + \dots \right] \end{aligned}$$

As  $|x| \rightarrow \infty$ ,  $\frac{1}{x^N} \rightarrow 0$ , So for large  $|x|$ ,  $P_N \approx a_N x^N$ .

Hence if  $a_N > 0$  WLOG, then as  $x \rightarrow \pm\infty$ ,  $P_{2N}(x) \rightarrow +\infty$  and  $P_{2N+1} \rightarrow \pm\infty$



## 1.5 Rational Functions

**Definition.** A function of the form  $f(x) = \frac{P(x)}{Q(x)}$ , where  $P(x)$  and  $Q(x)$  are polynomials is called a *rational function*.

Lecture 5

We *could* require that the order of  $P <$  order of  $Q$ . If this doesn't happen, we can use polynomial division to write:

$$\frac{P}{Q} = R(x) + \frac{S(x)}{Q(x)} \quad \text{where } R \text{ and } S \text{ are also polynomial}$$

**Example 1.15.**

$$\begin{aligned} \frac{x^2 + x}{x - 1} &= \frac{x^2 - x + 2x}{x - 1} \\ &= x + \frac{2x}{x - 1} \\ &= x + \frac{2x - 2 + 2}{x - 1} \\ &= (x + 2) + \frac{2}{x - 1} \end{aligned}$$

We could also require that  $P$  and  $Q$  have no common factors i.e.  $\nexists \alpha : P(\alpha) = 0 = Q(\alpha)$

Let's do this! (for simplicity). Any zero of  $Q(x)$  is then a *singularity* or *pole* or infinity of  $\frac{P}{Q}$  and is important. This behaviour is illustrated by...

### Partial Fraction Decomposition

Suppose  $Q$  has degree  $N$ , with no repeated root, i.e.

$$Q = \lambda(x - r_1)(x - r_2) \dots (x - r_N)$$

Where  $\lambda \neq 0, r_i \neq r = j$  unless  $i = j$  and  $r_i \in \mathbb{C}$

Then we can write:

$$\frac{P}{Q} = \frac{A_1}{x - r_1} + \frac{A_2}{x - r_2} + \dots + \frac{A_N}{x - r_N} + \overbrace{R(x)}^{\text{if needed}}$$

We can easily find  $A_i$  by multiplying through  $Q$ :

$$\begin{aligned} P &= \frac{A_1 Q}{x - r_1} + \frac{A_2 Q}{x - r_2} + \cdots + \frac{A_N Q}{x - r_N} \\ &= A_1 \lambda(x - r_2)(x - r_3) \cdots (x - r_N) + \frac{A_2 Q}{x - r_2} + \cdots + \frac{A_N Q}{x - r_N} \quad (*) \end{aligned}$$

Putting  $x = r_1$  into (\*);  $Q(r_1) = 0$ , so:

$$P(r_1) = A_1 \lambda(r_1 - r_2)(r_1 - r_3) \cdots (r_1 - r_N)$$

Using the product rule for differentiation, we also now have:

$$Q'(x) = \lambda[(x - r_2) \cdots (x - r_N)] + \lambda(x - r_1)[\text{a load of stuff}]$$

So

$$Q'(r_1) = \lambda(r_1 - r_2)(r_1 - r_3) \cdots (r_1 - r_N)$$

hence  $P(r_1) = A_1 Q'(r_1)$  or  $A_1 = \frac{P(r_1)}{Q'(r_1)}$

So obviously

$$A_i = \frac{P(r_i)}{Q'(r_i)} \quad i = 1, 2, \dots, N$$

*What could go wrong?*

(a) *What if (some of) the roots are complex?*

Algebra still works. But for some purposes we may prefer to keep things real.

$$\begin{aligned} \text{e.g. } \frac{3}{x^3 + 1} &= \frac{3}{(x + 1)(x^2 - x + 1)} = \frac{3}{(x + 1)(x - \omega)(x + \omega^*)} \\ &= \frac{A_1}{x + 1} + \frac{A_2}{x - \omega} + \frac{A_3}{x - \omega^*} \end{aligned}$$

Using the formula we obtained for  $A_i$ , we get  $A_1 = 1, A_2 = -\omega, A_3 = -\omega^*$

$$\begin{aligned} \frac{3}{x^3 + 1} &= \frac{1}{x + 1} - \frac{\omega}{x - \omega} - \frac{\omega^*}{x - \omega^*} \\ &= \frac{1}{x + 1} - \frac{x - 2}{x^2 - x + 1} \end{aligned}$$

Alternative Partial Fraction Form for real polynomials:

$$\frac{P}{Q} = R + \frac{A_1}{(x - r_1)} + \cdots + \underbrace{\frac{Cx + D}{Cx^2 + \delta x + \delta}}_{\text{for complex roots}} + \cdots$$

(b) What if there are repeated roots?

e.g.

$$Q(x) = \lambda(x - r_1)^2(x - r_3)(x - r_4) \dots (x - r_N)$$

Then

$$\frac{P}{Q} = R + \frac{A_1}{(x - r_1)^2} + \frac{B}{(x - r_1)} + \frac{A_3}{(x - r_3)} + \dots + \frac{A_N}{(x - r_N)}$$

Sometimes it's easiest to manipulate the numerator:

**Example 1.16.**

$$\frac{x}{(x + 2)^2(x + 3)}$$

*Example cancelled due to laziness of lecturer.*

## Use of Partial Fractions

Calculus and curve plotting. Every rational function can be integrated in terms of simple functions. Also useful when differentiating many times:

**Example 1.17.**

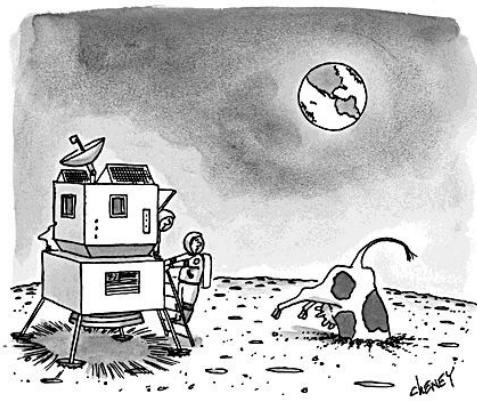
$$f = \frac{x + 3}{x^2 + 4x + 3}$$

We have  $P = x + 1$ ,  $Q' = 2x + 4$ , so  $A_1 = \frac{-1}{-2}$ ,  $A_2 = \frac{1}{2}$ :

$$\begin{aligned} f &= \frac{\frac{-1}{-2}}{x + 3} + \frac{\frac{1}{2}}{x + 1} \\ &= \frac{1}{2} \left[ \frac{1}{x + 3} + \frac{1}{x + 1} \right] \end{aligned}$$

So differentiating:

$$\begin{aligned} f' &= \frac{1}{2} \left( -\frac{1}{(x + 3)^2} - \frac{1}{(x + 1)^2} \right) \\ f'' &= \frac{1}{2} \left( \frac{2}{(x + 3)^3} + \frac{2}{(x + 1)^3} \right) \\ f^{(n)} &= \frac{1}{2} \left[ \frac{1}{(x + 3)^{n+1}} + \frac{1}{(x + 1)^{n+1}} \right] (-1)^n n! \end{aligned}$$



## 2 Infinite Series

**Definition.** An *infinite series* is a function of the form  $f(x) = \sum_{n=0}^{\infty} a_n x^n$  Lecture 6

We will assume for now that the infinite sum converges (i.e. tends to a limit) for at least some values of  $x$ . We will also assume we can manipulate infinite series sensibly. The  $a_n$  are called coefficients (may be  $\in \mathbb{C}$ )

### 2.1 The Exponential Function

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

In fact this series converges  $\forall x$ .

Forget everything we know about  $e^x$  for now... *What can we deduce about  $f(x)$ ?*

If  $x > 0$ ,  $f(x) > 1$  by inspection, and also  $x$  increases as  $f(x)$  increases. Question 6 on the problem sheets proved that  $\forall x, y, f(x)f(y) = f(x+y)$ .

Setting  $y = -x$ , yields  $f(-x) = \frac{1}{f(x)}$ , which tells us about when  $0 < x < 1$ .  $f(x) \rightarrow 0$ , as  $x \rightarrow -\infty$  etc. It follows that  $f(x)$  has an inverse function  $g(x)$  whose domain is  $(0, \infty)$  and range  $(-\infty, \infty)$ , so we know that  $x = g(f(x)) \forall x$  and  $x = f(g(x)) x > 0$ .

Now consider

$$\begin{aligned} x^2 &= x.x = f(g(x)) \cdot f(g(x)) \\ &= f(g(x) + g(x)) \quad [\text{By } f(x)f(y) = f(x+y)] \\ &= f(2g(x)) \end{aligned}$$

Clearly induction  $x^n = f(ng(x))$  for  $n \in \mathbb{N}$

**Definition.**  $x^\alpha = f(\alpha g(x)) = e^{\alpha \log x}$  for  $x > 0$ , any arbitrary  $\alpha$   
 $a^x = f(x g(a))$  for  $a > 0$ , any arbitrary  $x$

From the definition of  $f$ ,  $a^x = 1 + x(g(a)) + \frac{1}{2}[xg(a)]^2 + \frac{1}{3!}[xg(a)]^3 + \dots$

Choose  $a$  such that  $g(a) = 1$ , then we have:

$$a^x = 1 + 1 + \frac{1}{2} + \frac{1}{3!} + \dots = 2.718281828459\dots$$

Let's call this value (...wait for it),  $e$

Then  $g(e) = 1$ , so

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

From now we can use all the properties of the exponential function,

e.g.  $e^x e^y = e^{x+y}$ ,  $(e^a)^b = e^{ab}$ ,  $e^0 = 1$ .

Calling  $g(x) = \log(x) = \ln(x)$ , we have the usual properties which follows from the first problem sheet:

- $\log(uv) = \log(u) + \log(v)$
- $\log(1) = 0$
- “ $\log(0) = -\infty$ ”
- $\log(\frac{u}{v}) = \log(u) - \log(v)$
- $\log(a^b) = b \log a$
- $a^b = e^{b \log a}$

(We will not consider logarithms to different bases)

$e^x$  is defined  $x \in R$ , what if  $x$  is complex or purely imaginary?

**Definition.** Write  $x = i\theta, \theta \in R$ , then define  $e^{i\theta}$  to be:

$$\begin{aligned} f(i\theta) &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= 1 + i\theta + \frac{1}{2}(i\theta)^2 + \frac{1}{3!}(i\theta)^3 + \frac{1}{4!}(i\theta)^4 + \dots \\ &= \left(1 - \frac{1}{2}\theta^2 + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots\right) \end{aligned}$$

Now define

$$\cos(\theta) = \left(1 - \frac{1}{2}\theta^2 + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right)$$

and

$$\sin(\theta) = \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots\right)$$

We then have:

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

A better proof that  $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$  uses complex numbers, and the fact that: Lecture 7

$$\exp(x)\exp(y) \equiv \exp(x+y) \quad (*)$$

*Proof.* Consider  $\exp(i\theta)\exp(i\phi) \equiv \exp(i\theta + i\phi)$

$$\implies [\cos(\theta) + i\sin(\theta)][\cos(\phi) + i\sin(\phi)] \equiv \cos(\theta + \phi) + i\sin(\theta + \phi)$$

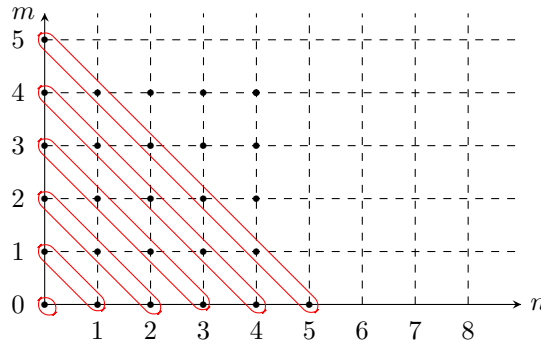
Expanding and equating the real parts gives required result. (Much easier than the series manipulation on handout 1!) ■

Now we prove identity (\*)...

**Proposition 2.1.**  $\exp(x) \exp(y) \equiv \exp(x + y)$

*Proof.* Consider  $\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ , then  $\exp(y) = \sum_{m=0}^{\infty} \frac{y^m}{m!}$

Then  $\exp(x) \exp(y) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{x^n}{n!} \frac{y^m}{m!}$



Assume it does not matter in which order we add up all the terms i.e. we can add the diagonals  $m + n = p$ . Indeed counting the terms diagonally, writing  $m + n = p$ :

$$\begin{aligned}
 \exp(x) \exp(y) &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{x^n}{n!} \frac{y^m}{m!} = \sum_{p=0}^{\infty} \sum_{n=0}^p \left( \frac{x^n y^{p-n}}{p!} \right) \\
 &= \sum_{p=0}^{\infty} \sum_{n=0}^p \left( \frac{{}^p C_n x^n y^{p-n}}{p!} \right) \\
 &= \sum_{p=0}^{\infty} \frac{1}{p!} \sum_{n=0}^p {}^p C_n x^n y^{p-n} \\
 &= \sum_{p=0}^{\infty} \frac{(x+y)^p}{p!} = \exp(x+y) \quad \blacksquare
 \end{aligned}$$

We now have  $\cos$  and  $\sin$ . You will prove on Sheet 2 Question 1 that they are  $2\pi$ -periodic and all the trigonometric formulae that follow, i.e.  $\cos(A+B)$  etc. *Remember them, or be able to derive them in 15 seconds!*

## 2.2 Other Infinite Series

There is no infinite series for  $\log(x)$ , i.e.

$$\log(x) \neq \sum_{n=0}^{\infty} a_n x^n \text{ for any } a_n$$

(try putting  $x = 0$ , and you would get “ $-\infty = a_0$ ”). But there one for  $\log(1+x)$ :

$$\log(1+x) = 0 + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 + \dots = \sum_{n=0}^{\infty} x^n (-1)^n, \quad |x| < 1$$

This is an example of the geometric series  $a+ar+ar^2+\dots = \frac{a}{1-r}$ ,  $|r| < 1$ . N.B. We could integrate the series for  $\frac{1}{1+x}$  to obtain  $\log(1+x)+c = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$  assuming integrating term by term is allowed. Letting  $x = 0 \implies c = 0$ .

## The Binomial Series

The Geometric series is a special case of the Binomial Series:

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-p)}{(p+1)!} x^{p+1} + \dots \quad (\alpha \in \mathbb{R})$$

This series converges provided  $|x| < 1$ . Note, if  $\alpha \in \mathbb{N}$ , then eventually the coefficients become zero and the power series *terminates* as a polynomial of degree  $N$ . The series then becomes the binomial theorem:  $(1+x)^n = \sum_{k=0}^n {}^nC_k x^k$ .

Proof by:

- (i) Leave it to M1F
- (ii) Induction

## 2.3 Hyperbolic Functions

**Definition.** We define

$$\cosh(x) = \text{Even part of } (e^x)$$

$$\sinh(x) = \text{Odd part of } (e^x)$$

i.e.

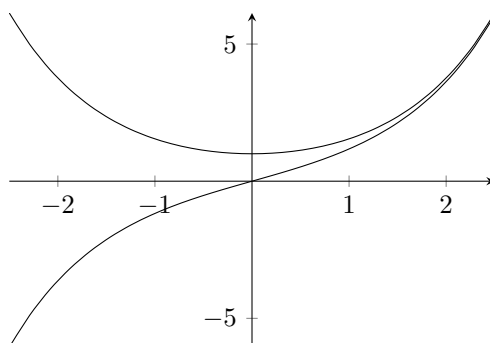
$$\cosh(x) \equiv \frac{1}{2}(e^x + e^{-x}), \quad \sinh(x) \equiv \frac{1}{2}(e^x - e^{-x})$$

These obviously have the series:

$$\cosh(x) = 1 + \frac{1}{2}x^2 + \frac{1}{4!}x^4 + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$$

(removing the odd powers from  $\exp(x)$ ), and

$$\sinh(x) = x + \frac{x^3}{6} + \frac{x^5}{5!} + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$



## Inverse Hyperbolic Functions

Suppose

$$y = \sinh x = \frac{e^x - e^{-x}}{2}$$

Solve for  $x$ . How? Write  $u = e^x$ :

$$\begin{aligned} 2y &= u - \frac{1}{u} \\ u^2 - 2uy - 1 &= 0 \\ u &= y \pm \sqrt{1 + y^2} \end{aligned}$$

Which root do we take?

$ue^x > 0 \forall x$ , and  $y - \sqrt{1 + y^2} < 0$  (if  $y > 0$ ), so we take the  $+$  sign.

$$e^x = y + \sqrt{1 + y^2}$$

So

$$\begin{aligned} x &= \log[y + \sqrt{1 + y^2}] \\ &\equiv \sinh^{-1} y \equiv \operatorname{arcsinh}(y) \end{aligned}$$

What about  $\cosh^{-1}$ ?

Write

$$y = \frac{1}{2}(e^x + e^{-x})$$

Letting  $u = e^x$  we have

$$u^2 - 2uy + 1 = 0$$

$$\implies u = y \pm \sqrt{y^2 - 1}$$

Which root do we take? Either is possible depending on the domain of  $\cosh x$ . Assume  $x \geq 0 \implies e^x \geq 1$ . Again, we need the + sign (for larger root), so

$$\begin{aligned} x &= \log[y + \sqrt{y^2 - 1}] \\ &= \cosh^{-1} y \equiv \operatorname{arccosh}(y) \end{aligned}$$

Note: taking the minus sign would give “ $-x$ ” instead. Indeed:

$$\begin{aligned} &\log(y + \sqrt{y^2 - 1}) + \log(y - \sqrt{y^2 - 1}) \\ &= \log[(y + \sqrt{y^2 - 1})(y - \sqrt{y^2 - 1})] \\ &= \log[y^2 - (\sqrt{y^2 - 1})^2] \\ &= \log[y^2 - (y^2 - 1)] = \log(1) = 0 \end{aligned}$$

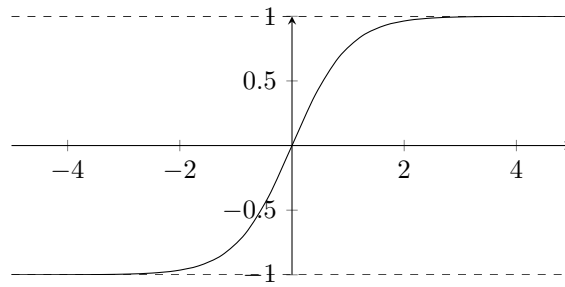
So our inverse hyperbolic functions are

$$\begin{aligned} \sinh^{-1}(x) &= \log(x + \sqrt{x^2 + 1}) \\ \cosh^{-1}(x) &= \log(x + \sqrt{x^2 - 1}) \end{aligned}$$

Note:  $\cosh^2 - \sinh^2 \equiv 1$ . There is much similarity with trigonometric functions.

See Problem Sheet 2 for  $\tanh x$  and  $\tanh^{-1} x$ .

$$\tanh x \equiv \frac{\sinh x}{\cosh x}$$



The  $\tanh$  function is very useful for switching between  $-1$  and  $1$  smoothly.

## 2.4 Expanding other functions as power series

We can use the functions we've defined to express many others as power series.

**Example 2.2.** Find the Power Series expansion for  $(1 + x^2) \cosh x$

$$\begin{aligned}
 (1 + x^2) \cosh x &= (1 + x^2) \left( 1 + \frac{x^2}{2} + \frac{x^4}{24} + \frac{x^6}{6!} + \dots \right) \\
 &= 1 + \frac{x^2}{2} + \frac{x^4}{24} + \mathcal{O}(x^6) \\
 &\quad + x^2 + \frac{x^4}{2} + \mathcal{O}(x^6) \\
 &= 1 + \frac{3x^2}{2} + \frac{13}{24}x^4 + \mathcal{O}(x^6)
 \end{aligned}$$

Note:  $\mathcal{O}(x^6)$  means terms at least as small as  $cx^6$  as  $x \rightarrow \infty$

**Example 2.3.** Find the Power Series expansion for

$$\frac{\sqrt{1 + \frac{x^2}{2}}}{\cos x}$$

Use the binomial theorem for expanding the numerator:

$$\begin{aligned}
 \sqrt{1 + \frac{x^2}{2}} &= \left( 1 + \frac{x^2}{2} \right)^{1/2} \\
 &= 1 + \frac{1}{2} \left( \frac{x^2}{2} \right) + \frac{(\frac{1}{2})(-\frac{1}{2})}{2} \left( \frac{x^2}{2} \right)^2 + \frac{(\frac{1}{2})(-\frac{1}{2})(-\frac{3}{2})}{3!} \left( \frac{x^2}{2} \right)^3 + \dots \\
 &= 1 + \frac{x^2}{4} - \frac{x^4}{32} + \mathcal{O}(x^6)
 \end{aligned}$$

Now

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4} + \mathcal{O}(x^6)$$

So how do we deal with

$$\begin{aligned}
 \frac{1}{\cos x} &= \frac{1}{1 - \frac{x^2}{2} + \frac{x^4}{4} + \mathcal{O}(x^6)} \\
 &= \frac{1}{1 - T}
 \end{aligned}$$

Define  $T = \frac{x^2}{2} - \frac{x^4}{4!} + \mathcal{O}(x^6)$ .

$$\begin{aligned}\frac{1}{1-T} &= 1 + T + T^2 + \mathcal{O}(T^3) \\ \frac{1}{\cos x} &= 1 + \left(\frac{x^2}{2} - \frac{x^4}{24}\right) + \left(\frac{x^2}{2} - \frac{x^4}{24}\right)^2 + \mathcal{O}(x^6) \\ &= 1 + \frac{x^2}{2} - \frac{x^4}{24} + \left(\frac{x^2}{2}\right)^2 + \mathcal{O}(x^6) \\ &= 1 + \frac{x^2}{2} + \frac{5x^4}{24}\end{aligned}$$

Putting them together:

$$\begin{aligned}\frac{\sqrt{1+\frac{x^2}{2}}}{\cos x} &= \left(1 + \frac{x^2}{4} - \frac{x^4}{32}\right) \left(1 + \frac{x^2}{2} + \frac{5x^4}{24}\right) + \mathcal{O}(x^6) \\ &= 1 + x^2 \left[\frac{1}{4} - \frac{1}{2}\right] + x^4 \left[1 \cdot \frac{5}{24} + \frac{1}{4} \cdot \frac{1}{2} - \frac{1}{32} \cdot 1\right] \\ &= 1 + \frac{3x^2}{4} + \frac{x^4}{96} [20 + 12 - 3] + \mathcal{O}(x^6) \\ &= 1 + \frac{3x^2}{4} + \frac{29x^4}{96} + \mathcal{O}(x^6)\end{aligned}$$

**Example 2.4.** A “simpler” example: Find the power series for

$$\frac{1}{1+e^x}$$

Try letting  $T = e^x$ :

$$\begin{aligned}\frac{1}{1+T} &= 1 - T + T^2 - T^3 + T^4 - T^5 + T^6 - T^7 + \dots \\ &= -(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots) \\ &\quad + (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)^2 \\ &\quad - (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)^3 \\ &\quad + (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)^4\end{aligned}$$

We would then try gathering the constant terms, the  $x$  terms together etc. But then we end up with  $-1 + 1 - 1 + \dots$  and  $x + x + x + \dots$ , which goes off to infinity!

*What's gone wrong?*

Suppose  $x$  is very small, then  $T = 1$  which is still large, so there is no justification for neglecting small powers of  $T$ ... we end up having to include all powers of  $T$  - for which we are DOOMED.

### Theorem 2.5: Golden Rule

Never ever ever expand in a quantity you are not prepared to treat as small.

**Example 2.6.** We broke the Golden Rule, so start Example 2.4 again:

Write  $T = e^x - 1$ .

$$\begin{aligned}
 \frac{1}{1+e^x} &= \frac{1}{1+1+T} \\
 &= \frac{1}{2+T} \\
 &= \frac{1}{2} \left(1 + \frac{T}{2}\right)^{-1} \\
 &= \frac{1}{2} \left(1 - \frac{T}{2} + \left(\frac{T}{2}\right)^2\right) + \dots \\
 &= \frac{1}{2} - \frac{T}{4} + \frac{T^2}{8} + \dots \\
 &= \frac{1}{2} - \frac{1}{4} \left(x + \frac{x^2}{2} + \mathcal{O}(x^3)\right) + \frac{1}{8} \left(x + \frac{x^2}{2}\right)^2 + \mathcal{O}(x^3) \\
 &= \frac{1}{2} - \frac{1}{4}x - \frac{1}{8}x^3 + \frac{1}{8}x^2 + \mathcal{O}(x^3) \\
 &= \frac{1}{2} - \frac{1}{4}x + \mathcal{O}(x^3)
 \end{aligned}$$

## 2.5 Maclaurin Series

*What Kind of functions have a power series?*

Lecture 9

i.e. When can we write  $f(x) = a_0 + a_1x + a_2x^2 + \dots a_nx^n = \sum_{n=0}^{\infty} a_nx^n$ ?

Note that if this is true then  $f(0) = a_0$ . So if  $f(x)$  is *differentiable* (see later) and it is legitimate to differentiate an infinite series term by term, then

$$f'(x) = a_1 + 2a_2x + \dots na_nx^{n-1} = \sum_{n=0}^{\infty} na_nx^{n-1}$$

Now we put  $x = 0$ , to get  $f'(0) = a_1$ .

In general, if the function  $f(x)$  can be differentiated  $r$  times (and so can the

series), then

$$f^{(r)}(0) = a_r r! x^0 + 0 + 0 + 0 + \dots \implies a_r = \frac{f^{(r)}(0)}{r!}$$

(if  $n < r$  in sum, one of the prefactors of  $x^{n-r}$  is 0. If  $n > r$ ,  $x^{n-r}$  is 0, when  $x = 0$ . So only one term,  $n = r$ , remains.) So formally,

**Definition.** if the function has a power series then we expect the *Maclaurin Series* (or the Taylor series about  $x = 0$ ) to be:

$$f(x) = f(0) + x f'(0) + \frac{x^2 f''(0)}{2!} + \dots + \frac{x^r f^{(r)}(0)}{r!} = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

We suspect therefore, the only those functions with an arbitrary number of derivatives of  $x = 0$  have the power series expansion.

So  $\log(x)$ ,  $x^{3.1}$ ,  $\sin(x^{\frac{1}{2}})$ , or  $|x^3|$  do not have series expansions. There are some functions for which the power series exists but converges to a different function. Such functions are called *non-analytical* functions. e.g.

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

We shall not worry about such functions anymore.

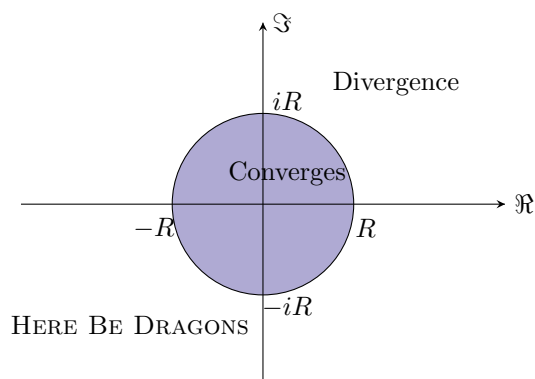
So a Maclaurin series exists  $\iff f^{(n)}(0)$  exists  $\forall n$ . e.g.  $f(x) = e^{-x} \sin(2x) = 2x - 2x^2 + \mathcal{O}(x^3)$

## 2.6 Radius of Convergence

**Definition.** For any power series  $\sum_{n=0}^{\infty} a_n x^n$  ( $a_n, x \in \mathbb{C}$ ),  $\exists \mathbb{R}$  such that:

- if  $|x| < R$  series converges
- if  $|x| > R$  series does not converge
- if  $|x| = R$  anything can happen
- if  $R = 0$  series converges only for  $x = 0$
- if “ $R = \infty$ ” series converges  $\forall x$

$R$  is called the *radius of convergence*



**Example 2.7.**  $(x + \alpha)^\alpha$   $a \neq 0 \in \mathbb{R}, \alpha \in \mathbb{R}$

$$\implies a^\alpha \left(1 + \frac{x}{a}\right)^\alpha$$

assuming  $a > 0$  (otherwise if  $a < 0$ , write  $a = -b, (x - b)^\alpha = b^\alpha \left(\frac{x}{b} - 1\right)^\alpha$ )

$$= (1 + t)^\alpha$$

we “know” this requires  $|t| < 1 \implies \left|\frac{x}{a}\right| < 1 \implies |x| < |a|$ , so  $\boxed{R = |a|}$ .

*What limits the circle of convergence?*

In practice, the series converges in as big a circle as it can i.e. until it reaches a singular point. e.g.  $\frac{1}{x^2+1}$  is singular (infinite) when  $x = i$ , so  $R \not> 1$  (else series would have to converge at  $x = i$ , but it can't as function is infinite there), so  $R = 1$  for this function.

He spent Lecture 10 going through his handout on Analysis. This basically summarises M1P1 ANALYSIS I, the most important results of which are:

Lecture 10

**Definition** (Convergence). We say  $u_n \rightarrow u$  if  $\forall n$  sufficiently large,  $|u_n - u|$  is arbitrarily small. Define a real number  $b \in \mathbb{R}$  to be arbitrarily small if it is smaller than any  $\epsilon > 0$ , i.e.  $\forall \epsilon > 0, |b| < \epsilon$ .

### Theorem 2.8: Comparison Test

Suppose  $a_n \geq b_n \geq 0$ . Then

$$\sum_{n=0}^{\infty} a_n \text{ convergent} \implies \sum_{n=0}^{\infty} b_n \text{ convergent}$$

**Theorem 2.9: Ratio Test**

If  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = l$ , then the series

$$\sum_{n=0}^{\infty} a_n \begin{cases} \text{converges if} & l < 1 \\ \text{diverges if} & l > 1 \\ \text{uncertain if} & l = 1 \end{cases}$$

We can use the ratio test to determine the radius of convergence for a power series. Applying it to  $\sum_{n=0}^{\infty} a_n x^n$ , we have:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} x^{n+1}}{a_n x^n} \right| = \frac{|x|}{R}$$

Then by the ratio test, the power series converges if  $|x| < R$ , diverges if  $|x| > R$  and may do either if  $|x| = R$ . So if the test works,  $R$  is in fact the *radius of convergence*.

Recall the Maclaurin series:

Lecture 11

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)x^n}{n!}$$

We can derive the *Taylor series* from the Maclaurin series in a few steps.

Define a new function

$$\begin{aligned} g(y+h) &= f(h) \text{ for some constant } y \\ f'(0) &= g'(y) \\ f''(0) &= g''(y-h) \text{ etc.} \end{aligned}$$

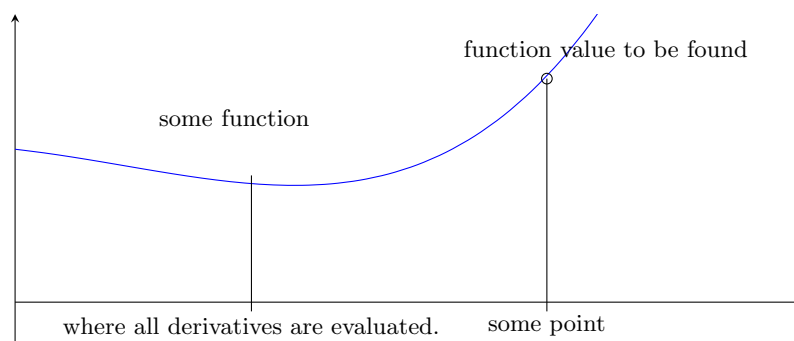
So we get

$$g(y+h) = \sum_{n=0}^{\infty} \frac{g^{(n)}(y)h^n}{n!}$$

Relabelling  $y \rightarrow x$  and  $g \rightarrow f$ , we get the Taylor series:

**Definition.** The *Taylor series* is

$$f(x+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)h^n}{n!}$$



### Finding the Radii of Convergence

**Example 2.10.**

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

To find the radius of convergence, look at the ratio of two adjacent terms.

$$\begin{aligned} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| &= \frac{|x|n!}{(n+1)!} \\ &= \frac{|x|}{n+1} \end{aligned}$$

Now take the limit as  $n \rightarrow \infty$ . Obviously this tends to 0. i.e.

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 0 < 1$$

So series converges for all  $x$  by the ratio test, so “ $R = \infty$ ”.

**Example 2.11.** Find the radius of convergence for

$$f(x) = \tan^{-1}(x) = \arctan(x)$$

Differentiating:

$$\begin{aligned} f'(x) &= \frac{1}{1+x^2} \text{ (see later if necessary!)} \\ &= 1 - x^2 + x^4 - x^6 + x^8 - \dots \end{aligned}$$

So assume

$$f(x) = C + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + \frac{x^{2n+1}}{2n+1}(-1)^n + \dots$$

When  $x = 0$ ,  $\tan^{-1} = 0 \implies C = 0$ , so we have the power series for the

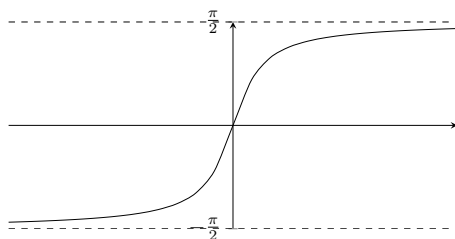
function:

$$f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

To find the radius of convergence, look at

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2n+3} / (2n+3)}{(-1)^n x^{2n+1} / (2n+1)} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(-1) x^2 (2n+1)}{(2n+3)} \right| \\ &= |x^2| \lim_{n \rightarrow \infty} \left| \frac{2n+1}{2n+3} \right| \\ &= |x^2| \lim_{n \rightarrow \infty} \left| \frac{1 + \frac{1}{2n}}{1 + \frac{3}{2n}} \right| \\ &= |x^2| < 1 \iff |x| < 1 \implies R = 1 \end{aligned}$$

Why is  $R = 1$ ? The function is perfectly well behaved in  $\mathbb{R}$ :



But in the complex plane this function is not nicely behaved - there is a singularity when  $x = i$ , so  $R \not\geq 1$ .

**Example 2.12.** Find  $R$  for

$$\sum_{n=0}^{\infty} \frac{x^2 n^2}{2^n (n+1)}$$

Look at

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{2^{n+1}} \frac{(n+1)^2}{(n+2)}}{\frac{x^n n^2}{2^n (n+1)}} \right| \\ &= \frac{|x|}{2} \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{n^2 (n+2)} \right| \\ &= \frac{|x|}{2} < 1 \iff |x| < 2 \end{aligned}$$

So  $R = 2$ .

To find  $R$ , we need to be able to find limits. This can often be hard. Normally

$$\lim_{x \rightarrow a} f(x) = f(a)$$

if  $f$  is continuous, but what if  $f(a)$  is hard to evaluate?

What is “ $\frac{0}{0}$ ”? It can be anything. Also troublesome is

$$\frac{\infty}{\infty}, 0 \times \infty, 0^\infty, \infty - \infty, 1^\infty$$

*How do we cope?*

**Example 2.13.** Find

$$\lim_{x \rightarrow \infty} x^{1/2}(\sqrt{x+1} - \sqrt{x+4})$$

Note

$$\begin{aligned} & \sqrt{x+1} - \sqrt{x+4} \\ &= (\sqrt{x+1} - \sqrt{x+4}) \frac{\sqrt{x+1} + \sqrt{x+4}}{\sqrt{x+1} + \sqrt{x+4}} \\ &= \frac{(x+1) - (x+4)}{\sqrt{x+1} + \sqrt{x+4}} \\ &= \frac{-3}{\sqrt{x+1} + \sqrt{x+4}} \end{aligned}$$

So our limit is now:

$$\begin{aligned} \lim_{x \rightarrow \infty} x^{1/2}(\sqrt{x+1} - \sqrt{x+4}) &= \lim_{x \rightarrow \infty} \frac{-3x^{1/2}}{\sqrt{x+1} + \sqrt{x+4}} \\ &= \lim_{x \rightarrow \infty} \left[ \frac{-3x^{1/2}}{x^{1/2}(1 + \frac{1}{x})^{1/2} + x^{1/2}(1 + \frac{4}{x})^{1/2}} \right] \\ &= \frac{-3}{2} \end{aligned}$$

**Example 2.14.** Find

$$\lim_{x \rightarrow 0} \left[ (\cosh(\sqrt{x}))^{1/2} \right]$$

This tends to  $1^\infty$ , which is a problem. But recalling that  $\cosh t = 1 +$

$\frac{1}{2}t^2 + \dots$ , we can consider the log:

$$\begin{aligned} \log \left[ (\cosh(\sqrt{x}))^{1/2} \right] &= \frac{1}{x} \log(\cosh \sqrt{x}) \\ &\approx \frac{1}{x} \log\left(1 + \frac{1}{2}(\sqrt{x})^2\right) \\ &= \frac{1}{x} \log\left(1 + \frac{1}{2}x\right) \\ &\approx \frac{1}{x} \left(\frac{1}{2}x\right) = \frac{1}{2} \end{aligned}$$

noting that  $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} + \dots$

So

$$\lim_{x \rightarrow 0} \left[ (\cosh(\sqrt{x}))^{1/2} \right] = e^{1/2}$$



"Now watch how I lift my tray table to its original and upright position. "

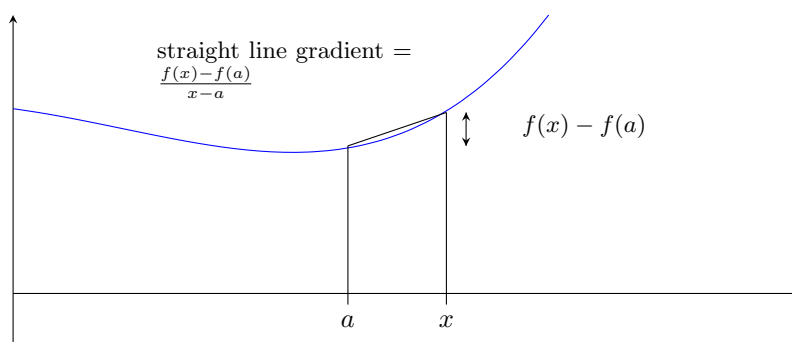
## 3 Differentiation

### 3.1 First Principles Differentiation

What is  $\lim_{x \rightarrow a} (f(x) - f(a))$  if  $f(x)$  is continuous? (= 0 obviously.)

Lecture 12

What about  $\lim_{x \rightarrow a} \left[ \frac{f(x) - f(a)}{x - a} \right]$ ?



We are taking the limit of the line joining  $(a, f(a))$  to  $(x, f(x))$ .

$$\frac{f(x) - f(a)}{x - a} \rightarrow \text{Gradient}$$

**Definition.** Then as  $x \rightarrow a$ , the limit tends to the gradient of the tangent. If the limit

$$\lim_{x \rightarrow a} \left[ \frac{f(x) - f(a)}{x - a} \right] \quad (*)$$

exists, we call it the “*derivative* of the function  $f(x)$  at  $a$ ”.

We can denote this limit  $(*)$  as either  $\frac{df}{dx}$  or  $f'(x)$ .

If in addition this limit  $(*)$  exists for every point on an interval  $c < x < d$  ( $x \in (c, d)$ ), we say  $f(x)$  is *differentiable* on the interval  $(c, d)$  and we have then defined a new function  $f'(x)$  on  $x \in (c, d)$ .

**Definition.** More commonly we define

$$f'(x) = \lim_{\epsilon \rightarrow 0} \left[ \frac{f(x + \epsilon) - f(x)}{\epsilon} \right]$$

N.B. This definition is equivalent to  $(*)$ , with  $\epsilon = x - a$ . (*Exercise:* Check this.)

Lets do some examples:

**Example 3.1.** Differentiate  $f(x) = x^2$  from first principles

$$\begin{aligned} f'(x) &= \lim_{\epsilon \rightarrow 0} \left[ \frac{(x + \epsilon)^2 - x^2}{\epsilon} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{x^2 + 2\epsilon x + \epsilon^2 - x^2}{\epsilon} \right] \\ &= \lim_{\epsilon \rightarrow 0} [2x + \epsilon] \\ &= 2x \end{aligned}$$

**Example 3.2.** Differentiate from first principles  $f(x) = x^{1/2}$

$$f'(x) = \lim_{\epsilon \rightarrow 0} \left[ \frac{(x + \epsilon)^{1/2} - x^{1/2}}{\epsilon} \right]$$

**Beware** of the temptation to use the binomial series as we haven't proved it!!

We use a trick:  $a^2 - b^2 = (a + b)(a - b)$ . Let  $a = (x + \epsilon)^{1/2}$  and  $b = x^{1/2}$ , so then:

$$\begin{aligned} f'(x) &= \lim_{\epsilon \rightarrow 0} \left[ \frac{(x + \epsilon)^{1/2} - x^{1/2}}{\epsilon} \cdot \frac{(x + \epsilon)^{1/2} + x^{1/2}}{(x + \epsilon)^{1/2} + x^{1/2}} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{((x + \epsilon)^{1/2})^2 - (x^{1/2})^2}{\epsilon((x + \epsilon)^{1/2} + x^{1/2})} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{(x + \epsilon) - x}{\epsilon((x + \epsilon)^{1/2} + x^{1/2})} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{\epsilon}{\epsilon((x + \epsilon)^{1/2} + x^{1/2})} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{1}{(x + \epsilon)^{1/2} + x^{1/2}} \right] \\ &= \frac{1}{x^{1/2} + x^{1/2}} \\ &= \frac{1}{2} \frac{1}{x^{1/2}} = \frac{1}{2} x^{-1/2} \end{aligned}$$

## Product Rule

Suppose we have  $f(x)$  and  $g(x)$  that are differentiable, i.e. their derivatives exists. We ask what is the derivative of their product  $f(x) \cdot g(x)$ ??

$$\begin{aligned}
\frac{d}{dx}(fg) &= \lim_{\epsilon \rightarrow 0} \left[ \frac{f(x+\epsilon)g(x+\epsilon) - f(x)g(x)}{\epsilon} \right] \\
&= \lim_{\epsilon \rightarrow 0} \left[ \frac{f(x+\epsilon)g(x+\epsilon) + 0 - f(x)g(x)}{\epsilon} \right] \\
\frac{d}{dx}(fg) &= \lim_{\epsilon \rightarrow 0} \left[ \frac{f(x+\epsilon)g(x+\epsilon) - f(x)g(x+\epsilon) + f(x)g(x+\epsilon) - f(x)g(x)}{\epsilon} \right] \\
&= \lim_{\epsilon \rightarrow 0} \left[ g(x+\epsilon) \left( \frac{f(x+\epsilon) - f(x)}{\epsilon} \right) + f(x) \left( \frac{g(x+\epsilon) - g(x)}{\epsilon} \right) \right] \\
&= \lim_{\epsilon \rightarrow 0} [g(x) + \epsilon] \lim_{\epsilon \rightarrow 0} \left( \frac{f(x+\epsilon) - f(x)}{\epsilon} \right) + \lim_{\epsilon \rightarrow 0} [f(x) + \epsilon] \lim_{\epsilon \rightarrow 0} \left( \frac{g(x+\epsilon) - g(x)}{\epsilon} \right) \\
&= g(x)f'(x) + f(x)g'(x)
\end{aligned}$$

We have shown:  $(fg)' = f'g + g'f$ .

### Chain Rule

Suppose again that  $f, g$  are differentiable in appropriate domains, then what is  $\frac{d}{dx}(f(g(x)))$ ?

By definition

$$\frac{d}{dx}(f(g)) = \lim_{\epsilon \rightarrow 0} \left[ \frac{f(g(x+\epsilon)) - f(g(x))}{\epsilon} \right]$$

*Trick:* We are going to multiply by 1 in a clever way by multiplying the top and bottom by  $g(x+\epsilon) - g(x)$ . [We know the answer will have  $g'$  in it, but we are not cheating... not really]

$$\begin{aligned}
\frac{d}{dx}(f(g)) &= \lim_{\epsilon \rightarrow 0} \left[ \frac{f(g(x+\epsilon)) - f(g(x))}{\epsilon} \cdot \frac{g(x+\epsilon) - g(x)}{g(x+\epsilon) - g(x)} \right] \\
&= \lim_{\epsilon \rightarrow 0} \left[ \frac{f(g(x+\epsilon)) - f(g(x))}{g(x+\epsilon) - g(x)} \right] \lim_{\epsilon \rightarrow 0} \left[ \frac{g(x+\epsilon) - g(x)}{\epsilon} \right]
\end{aligned}$$

*Trick:* Let  $\theta = g(x+\epsilon) - g(x) \implies g(x+\epsilon) = g(x) + \theta$ .

I claim that  $\lim_{\epsilon \rightarrow 0} \equiv \lim_{\theta \rightarrow 0}$ . So

$$\frac{d}{dx}(f(g)) = \lim_{\epsilon \rightarrow 0} \left[ \frac{g(x+\epsilon) - g(x)}{\epsilon} \right] \lim_{\theta \rightarrow 0} \left[ \frac{f(g(x) + \theta) - f(g(x))}{\theta} \right]$$

Let  $g(x) = u$ , so

$$\lim_{\theta \rightarrow 0} \left[ \frac{f(u + \theta) - f(u)}{\theta} \right] = \frac{df}{du}$$

Then we have

$$\boxed{\frac{d}{dx}(f(g)) = \frac{dg}{dx} \cdot \frac{df}{du}}$$

### Quotient Rule

What is  $\frac{d}{dx} \left( \frac{f(x)}{g(x)} \right)$ ?

We will use the product rule:  $\left( \frac{f}{g} \right)' = f' \left( \frac{1}{g} \right) + f \left( \frac{1}{g} \right)'$ .

Aside: What is  $\left( \frac{1}{x} \right)'$ ?

$$\begin{aligned} \lim_{h \rightarrow 0} \left[ \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \right] &= \lim_{h \rightarrow 0} \left[ \frac{x - (x+h)}{h(x+h)(x)} \right] \\ &= \lim_{h \rightarrow 0} \left[ \frac{-1}{(x+h)(x)} \right] \\ &= -\frac{1}{x^2} = -x^{-2}. \end{aligned}$$

Therefore  $\frac{d}{dx} \left( \frac{1}{g} \right) = \frac{d}{dg} \left( \frac{1}{g} \right) \frac{dg}{dx} = -\frac{1}{g^2} \cdot g'$ . Putting it all together, we find:

$$\begin{aligned} \left( \frac{f}{g} \right)' &= \frac{f'}{g} - \frac{fg'}{g^2} = \frac{f'g - fg'}{g^2} \\ \Rightarrow \boxed{\frac{d}{dx} \left( \frac{f(x)}{g(x)} \right) &= \frac{\frac{df}{dx}g - f\frac{dg}{dx}}{g^2}} \end{aligned}$$

With all these rules we can find the derivative to say:  $\log(1 + \sin(2^x + \sin(x)^{\cos(x)}))$ . The product rule and chain rule extend to more than two functions e.g.

Lecture 13

$$(fgh)' = f'(gh) + f(gh)' = f'(gh) + fg'h + (fg)h'$$

Similarly (by defining  $g(h(x)) \equiv k(x)$ ):

$$f[g(h[x])] = f[k(x)] = f'(k(x)) \cdot k'(x) = f'(g(h(x)))g'(h(x))h'(x)$$

It's easier to just remember that:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dt} \frac{dt}{d\Omega} \frac{d\Omega}{d\chi} \frac{d\chi}{d\xi} \frac{d\xi}{dx} \text{ etc.}$$

**Example 3.3.** What is  $\frac{d}{dx}(e^x)$ ?

$$\begin{aligned} &= \lim_{\epsilon \rightarrow 0} \left( \frac{e^{x+\epsilon} - e^x}{\epsilon} \right) = e^x \lim_{\epsilon \rightarrow 0} \left( \frac{e^\epsilon - 1}{\epsilon} \right) = e^x \lim_{\epsilon \rightarrow 0} \left( \frac{1 + \epsilon + \mathcal{O}(\epsilon^2) - 1}{\epsilon} \right) \\ &= e^x \lim_{\epsilon \rightarrow 0} [1 + \mathcal{O}(\epsilon)] \\ &= e^x \end{aligned}$$

*Exercise:* Show that  $\frac{dx}{dx} = 1$  from first principles

So  $1 = \frac{dx}{dx} = \frac{dx}{dy} \frac{dy}{dx}$  (using chain rule). i.e.  $\boxed{\frac{dy}{dx} = \frac{1}{dx/dy}}$

N.B. Next term you will meet partial derivatives,  $\frac{\partial u}{\partial x}$ ; The chain rule for partial differentiation is more complicated, and  $\frac{\partial u}{\partial x} \neq \frac{1}{\partial x / \partial u}$  necessarily.

If  $y = \log x$ , What is  $\frac{dy}{dx}$ ?

Write  $x = e^y \implies \frac{dx}{dy} = e^y \implies \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$

### Inverse Trigonometric Derivatives

What is  $\frac{d}{dx}(\sin^{-1} x)$ ? First we should find out what  $\frac{d}{dy}(\sin y)$  is:

$$\begin{aligned} \frac{dx}{dy} &= \frac{d}{dy}(\sin y) \\ &= \frac{d}{dy} \left[ y - \frac{y^3}{6} + \frac{y^5}{120} + \dots \frac{y^n}{n!} \right] \\ &= \left[ 1 - \frac{y^2}{2} + \frac{y^4}{24} + \dots \right] \\ &= \cos y \end{aligned}$$

Or we could use first principles:

$$\begin{aligned} \frac{d}{dy}[\sin y] &= \lim_{\epsilon \rightarrow 0} \left[ \frac{\sin(y + \epsilon) - \sin(y)}{\epsilon} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{\sin y \cos \epsilon + \sin \epsilon \cos y - \sin y}{\epsilon} \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{\sin \epsilon}{\epsilon} \cos y \right] + \lim_{\epsilon \rightarrow 0} \left[ \sin y \left( \frac{\cos \epsilon - 1}{\epsilon} \right) \right] \\ &= \lim_{\epsilon \rightarrow 0} \left[ \frac{\epsilon + \mathcal{O}(\epsilon^3)}{\epsilon} \right] \cos y + \sin y \lim_{\epsilon \rightarrow 0} \left[ \frac{1 - \frac{\epsilon^2}{2} + \mathcal{O}(\epsilon^4) - 1}{\epsilon} \right] \\ &= \cos y \end{aligned}$$

Similarly  $\frac{d}{dx}(\cos x) = -\sin x$ .

For  $\tan(x)$ , we have:

$$\begin{aligned}\frac{d}{dx}(\tan x) &= \left( \frac{\sin(x)}{\cos(x)} \right) = 1 + \frac{\sin^2 x}{\cot x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} \\ &= \sec^2 x\end{aligned}$$

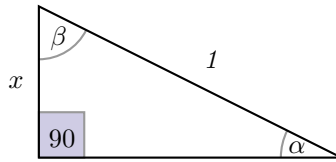
Return to  $\frac{d}{dx}(\sin^{-1} x)$ . Write  $x = \sin y$ , then  $\frac{dx}{dy} = \cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$ .

Hence

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}} = \frac{d}{dx}[\sin^{-1} x]}$$

*Exercise:* Show  $\frac{d}{dx}(\cos x) = -\frac{1}{\sqrt{1 - x^2}}$

Hence  $\frac{d}{dx}[\sin^{-1} x + \cos^{-1} x] = 0$ , which is clear from the right angled triangle:



Since  $\alpha = \sin^{-1} x$  and  $\beta = \cos^{-1} x$ ,  $\alpha + \beta = \frac{\pi}{2}$ , obviously  $\frac{d}{dx} \frac{\pi}{2} = 0$ .

## Logarithmic Derivative

$$\frac{d}{dx}(\log(u(x))) = \frac{d}{du} \log u \cdot \frac{du}{dx} = \frac{1}{u} u'$$

This is called a logarithmic derivative. Useful when performing integration e.g.

$$I = \int \frac{e^x + \cos x}{1 + e^x + \sin x} dx$$

Observe that  $u = 1 + e^x + \sin x$  and  $u' = e^x + \cos x$

$$\implies I = \log(1 + e^x + \sin x) + c$$

## Implicit Differentiation

If  $y$  is given implicitly in terms of  $x$  e.g.  $y + \tan y = x$ , we can differentiate the entire equation term by term with respect to  $x$ :

**Example 3.4.** Differentiate  $y + \tan y = x$ :

$$\begin{aligned}\frac{d}{dx}(y + \tan y) &= \frac{d}{dx}(x) = 1 \\ \implies \frac{dy}{dx} \frac{d}{dy}(y + \tan y) &= 1 \\ \implies \frac{dy}{dx}(1 + \sec^2 y) &= 1 \\ \implies \frac{dy}{dx} &= \frac{1}{\sec^2 y + 1} = \frac{1}{2 + \tan^2 y} = \frac{1}{2 + (x - y)^2}\end{aligned}$$

## Higher Derivatives

If  $f(x)$  is differentiable in  $(a, b) \iff f'(x)$  is defined on  $(a, b)$ . Maybe  $f'$  is also differentiable. If so, we write it as  $f''$  or  $\frac{d^2 f}{dx^2}$  or  $\left(\frac{d}{dx}\right)^2 f$ , but not ever  $\left(\frac{df}{dx}\right)^2$ .

Continuing, we can write the  $n$ 'th derivative (if it exists) as  $f^{(n)}(x)$  or  $\frac{d^n f}{dx^n}$  or  $\left(\frac{d}{dx}\right)^n f$ . These forms are useful for Taylor / Maclaurin series.

## 3.2 Leibniz' Rule

*How can we (easily) differentiate a product many times?*

Lecture 14

Suppose  $f$  and  $g$  are differentiable an arbitrarily number of times. What is  $(fg)'$ ? Use the product rule:

$$\begin{aligned}(fg)' &= f'g + fg' \\ (fg)'' &= (f'g + fg')' = (f'g)' + (fg')' = f''g + 2f'g' + fg'' \\ (fg)''' &= f'''g + 3f''g' + 3f'g'' + fg'''\end{aligned}$$

We spot a pattern, and make an inspired (but intelligent) guess:

$$(fg)^n = \sum_{r=0}^n \binom{n}{r} f^{(r)} g^{(n-r)} \quad \left. \vphantom{\sum_{r=0}^n} \right\} \text{Leibniz' Formula} \quad (*)$$

This is very similar to the binomial theorem:

$$(f + g)^n = \sum_{r=0}^n \binom{n}{r} f^r g^{n-r}$$

(which I had hoped would have been proved in M1F)

*Proof.* Use induction to prove (\*)

(A) Take  $n = 1$  :  $(fg)' = f'g + fg'$  by product rule, so true for  $n = 1$

(B) Assume (\*) holds when  $n = k$ , and try to prove it then holds for  $n = k + 1$ .  
Hence:

$$\begin{aligned}(fg)^{(k)} &= \sum_{r=0}^k \binom{k}{r} f^{(r)} g^{(k-r)}, \text{ and differentiate again:} \\ \Rightarrow (fg)^{(k+1)} &= \sum_{r=0}^k \binom{k}{r} \left[ f^{(r+1)} g^{(k-r)} + f^{(r)} g^{(k-r+1)} \right] \\ &= f^{(s)} g^{(k+1-s)} \left[ \binom{k}{s-1} + \binom{k}{s} \right]\end{aligned}$$

**Lemma 3.5.**  $\binom{k}{s-1} + \binom{k}{s} = \binom{k+1}{s}$

*Proof*

- A) Pester Alessio
- B) Pester Emma
- C) Use Pascal's Triangle
- D) Use Factorials

Assuming lemma, we obtain:

$$(fg)^{(k+1)} = \sum_{s=0}^{k+1} \binom{k+1}{s} f^{(s)} g^{(k+1-s)}$$

as required.

(C) Hence by induction (\*) holds for all natural  $\mathbb{N}$ . ■

Note: Leibniz is very useful is one of the functions  $\omega \log f$  is a polynomial, as then the high derivatives vanish (are zero)

**Example 3.6.** What is  $\frac{d}{dx}(x^2 \sin x)$ ?

Using Leibniz,  $f = \sin x, g = x^2$ :

$$\begin{aligned}\frac{dy}{dx} &= (\sin x)^{(100)} x^2 + \binom{100}{1} \sin x^{(99)} \cdot 2x + \binom{100}{2} \sin x^{(98)} \cdot 2 + 0 + 0 + \dots \\ &= \sin x [x^2 - 9900] - 200x \cos x\end{aligned}$$

**Example 3.7.** : What is the Maclaurin series for  $y = \sin^{-1} x$ ?

$$y' = \frac{1}{\sqrt{1-x^2}}, \quad y'' = x(1-x^2)^{-3/2}$$

$$\implies (1-x^2)y'' = xy'$$

Differentiate entire equation  $n$  times using Leibniz rule:

$$[(1-x^2)y'']^{(n)} = [xy']^{(n)} = xy^{(n+1)} + ny^{(n)}$$

Note that:

$$[(1-x^2)y'']^{(n)} = (1-x^2)y^{(n+2)} + n(2x)y^{(n+1)} + \frac{n(n-1)}{2}(-2)y^{(n)}$$

Hence

$$(1-x^2)y^{(n+2)} - (2n+1)xy^{(n+1)} - n^2y^{(n)} = 0$$

So at  $x = 0$ , we have:

$$y^{(n+1)}(0) = n^2y^{(n)}(0) \quad \forall n$$

Now  $y = \sin^{-1} x \implies y(0) = 0$ , and  $y' = \frac{1}{\sqrt{1-x^2}} \implies y'(0) = 1$ . This means all even derivatives are zero. Considering the odd derivatives:

$$y^{(3)}(0) = 1, y^{(5)}(0) = 3^2, y^{(7)}(0) = 5^2 \cdot 3^2$$

So

$$\sin^{-1} x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} (n-2)^2(n-4)^2 \dots 5^2 \times 3^2 \times 1$$

### 3.3 Four Theorems and an Example

#### Theorem 3.8

Suppose  $f(x)$  is continuous on  $[a, b]$  i.e.  $a \leq x \leq b$  and differentiable on  $(a, b)$  i.e.  $a < x < b$ . Then  $f(x)$  attains its maximum and minimum values somewhere in  $[a, b]$ . i.e.

$$\exists c \ a \leq c \leq b, \text{ s.t. } f(c) \geq f(x) \ \forall x \in [a, b]$$

Furthermore, either  $c = a$  or  $c = b$  or  $f'(c) = 0$ .

Lecture 15

#### Theorem 3.9: Rolle's Theorem

If  $f(a) = f(b) = 0$ , and  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then  $\exists$  a point s.t.  $f'(c) = 0$ , where  $a < c < b$ .

*Sketch Proof.* Choose  $c$  to be the maximum of  $f$  over  $[a, b]$ .

Consider

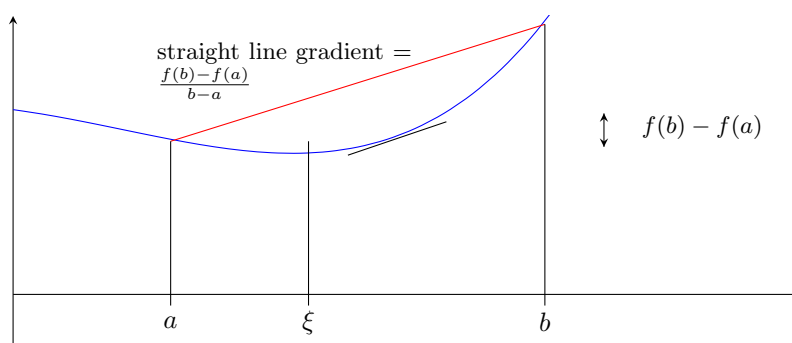
$$\frac{f(c + \epsilon) - f(c)}{\epsilon} \leq 0 \text{ where } \epsilon > 0$$

because  $f(c)$  is a maximum.

Consider

$$\frac{f(c) - f(c - \epsilon)}{\epsilon} \geq 0$$

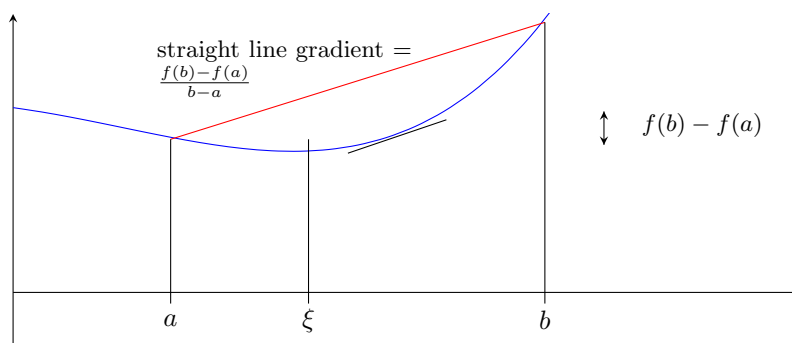
Then take the limit as  $\epsilon \rightarrow 0$ , both tend to  $f'(c)$  but one is  $\geq 0$ , and the other is  $\leq 0$ . ■



### Theorem 3.10: Mean Value Theorem

Suppose  $f$  is continuous in  $[a, b]$  and differentiable in  $(a, b)$ . Then  $\exists \xi \in (a, b)$  such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi)$$



Proof in two ways (assuming Rolle's Theorem)

- (i) Proof by Origami
- (ii) If you are unconvinced, proof by Maths.

*Proof.* Define a new function

$$g(x) = f(x) - f(a) - \left[ \frac{f(b) - f(a)}{b - a} \right] (x - a)$$

If  $g(a) = g(b) = 0$  and  $g$  is continuous and differentiable.

Now use Rolle's Theorem on  $g(x) \implies \exists \xi$  s.t.  $g'(\xi) = 0$  and  $a < \xi < b$ . Now  $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$ , so  $0 = g'(\xi) = f'(\xi) - \frac{f(b)-f(a)}{b-a}$ , or

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} \quad \blacksquare$$

### 3.4 De L'Hopital's Rule

#### Theorem 3.11: De L'Hopital's Rule

If  $f(x)$  and  $g(x)$  are differentiable in some interval about  $x = a$ , and  $f(a) = g(a)$ , and  $g'(a) \neq 0$  then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}$$

or

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Provided both limits exist [Second one used when  $g'(a) = 0$ ]

*Proof.* Use MVT. Assume  $x > a$ .  $\frac{f(x)-f(a)}{x-a} = f'(\xi)$  for some  $\xi \in (a, x)$ . Also  $\frac{g(x)-g(a)}{x-a} = g'(z)$  for some  $z \in (a, x)$ . Now  $f(a) = g(a) = 0$ . So

$$\frac{f(x)}{g(x)} = \frac{(x-a)f'(\xi)}{(x-a)g'(z)} = \frac{f'(\xi)}{g'(z)}$$

Then

$$\lim_{x \rightarrow a} \left[ \frac{f(x)}{g(x)} \right] = \lim_{x \rightarrow a} \left[ \frac{f'(\xi)}{g'(z)} \right]$$

Where  $\xi, z \in (a, x)$ . ■

**Example 3.12.** Find

$$\lim_{x \rightarrow 0} \left[ \frac{1 - \cos x}{x^2} \right]$$

We could also use power series for  $\cos(x)$ .

Using de L'Hopital:

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left[ \frac{(1 - \cos)' }{(x^2)'} \right] \\
 &= \lim_{x \rightarrow 0} \left[ \frac{\sin x}{2x} \right] \quad (\text{still "0/0"}) \\
 &= \lim_{x \rightarrow 0} \left[ \frac{\cos x}{2} \right] = \frac{1}{2}
 \end{aligned}$$

**Example 3.13.** Find

$$\lim_{x \rightarrow \pi/2} \left[ \frac{\cos x}{\log(\pi/2x)} \right]$$

Using de L'Hopital:

$$\begin{aligned}
 \lim_{x \rightarrow \pi/2} \left[ \frac{\cos x}{\log(\pi/2x)} \right] &= \lim_{x \rightarrow \pi/2} \left[ \frac{-\sin x}{-1/x} \right] \\
 &= \frac{-\sin \pi/2}{-1/(\pi/2)} \\
 &= 0
 \end{aligned}$$

**Example 3.14.** Find

Lecture 16

$$\lim_{x \rightarrow 2} \left[ \frac{(x-2) \log(x-1)}{\tan^2 \pi x} \right]$$

As  $x \rightarrow 2$ , the numerator  $\rightarrow 0$ , denominator  $\rightarrow 0$ . So we use Del'hop:  
[Notice you could replace  $\tan$  with  $\sin$  since  $\cos \rightarrow 1$  as  $n \rightarrow 2$ .]

$$\begin{aligned}
 \lim_{x \rightarrow 2} \left[ \frac{(x-2) \log(x-1)}{\tan^2 \pi x} \right] &= \lim_{n \rightarrow 2} \left[ \frac{\log(x-1) + \frac{x-2}{x-1}}{2 \tan \pi x \cdot \sec^2 \pi x \cdot \pi} \right] \\
 \lim_{n \rightarrow 2} \left[ \frac{\log(x-1) + \frac{x-2}{x-1}}{2 \tan \pi x \cdot \sec^2 \pi x \cdot \pi} \right] &= \frac{1}{2\pi \sec^2 2\pi} \lim_{x \rightarrow 2} \left[ \frac{\log(x-1) + \frac{(x-1)-1}{x-1}}{\tan \pi x} \right] \\
 &= \frac{1}{2\pi \cdot 1} \lim_{x \rightarrow 2} \left[ \frac{\frac{1}{x-1} + \frac{1}{(x-1)^2}}{\pi \sec^2 \pi x} \right] \\
 &= \frac{1}{2\pi} \left[ \frac{1+1}{\pi} \right] \\
 &= \frac{1}{\pi^2}
 \end{aligned}$$

Note: de L'Hopital's rule also works for expressions of the form " $\frac{\infty}{\infty}$ ", but we haven't justified this. If you believe this, then we can prove:

**Proposition 3.15.**

$$\lim_{x \rightarrow \infty} \left[ \frac{x^n}{e^{\alpha x}} \right] \quad (\alpha, n > 0)$$

i.e. Exponentials “beat” powers.

*Proof.* Using de L’Hopital’s rule:

$$\begin{aligned} \lim_{x \rightarrow \infty} \left[ \frac{x^n}{e^{\alpha x}} \right] &= \lim_{x \rightarrow \infty} \left[ \frac{nx^{n-1}}{\alpha e^{\alpha x}} \right] \\ &= \lim_{x \rightarrow \infty} \left[ \frac{n(n-1)x^{n-2}}{\alpha^2 e^{\alpha x}} \right] \\ &= 0 \end{aligned}$$

■

Is this proof OK?

*Alternative Proof.* Consider

$$\begin{aligned} 0 \leq \frac{x^n}{e^{\alpha x}} &= \frac{x^n}{1 + \alpha x + \frac{1}{2}\alpha^2 x^2 + \dots + \frac{1}{(n+1)!}\alpha^{n+1}x^{n+1}} \\ &< \frac{x^n}{\frac{1}{(n+1)!}\alpha^{n+1}x^{n+1}} \\ &= \frac{A}{x} \rightarrow 0 \text{ as } x \rightarrow \infty \end{aligned}$$

■

It follows (see Problem Sheet 2) that:

$$\lim_{x \rightarrow 0^+} x^\alpha (\log x) = 0$$

(where  $\alpha > 0$ )

\* Exponentials “beat” powers which “beats” logs in any struggle. \*

## Applications of the Mean Value Theorem

Recall MVT: If  $f$  is continuous and differentiable on  $[a, b]$ , then  $\exists \xi \in (a, b)$  such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi)$$

RHS is unknown. But if  $f'$  can be bounded in some way, we can estimate things well.

**Example 3.16.** What is  $\sin^{-1}(0.7)$ ?

Use the MVT. Let  $f(x) = \sin^{-1}(x)$ ,  $a = 0.7$ ,  $b = \frac{\sqrt{2}}{2} = 0.7071$ . Then  $\sin^{-1}(b) = \frac{\pi}{4}$ .

Also  $f'(x) = \frac{1}{\sqrt{1-x^2}}$ , so the MVT says:

$$\frac{\sin^{-1}(\frac{\sqrt{2}}{2}) - \sin^{-1}(0.7)}{\frac{\sqrt{2}}{2} - 0.7} = \frac{1}{\sqrt{1-\xi^2}}$$

For  $0.7 < \xi < 0.7071$ .

Now  $\frac{1}{\sqrt{1-x^2}}$  is an increasing function. So

$$\frac{1}{\sqrt{1-(0.7)^2}} < \frac{1}{\sqrt{1-\xi^2}} < \frac{1}{\sqrt{1-\frac{1}{2}}} = \sqrt{2}$$

$$\frac{\frac{\pi}{4} - \sin^{-1}(0.7)}{\frac{1}{\sqrt{2}} - 0.7} = \frac{1}{\sqrt{1-\xi^2}} < \sqrt{2}$$

$$\begin{aligned} \Rightarrow \frac{\pi}{4} - \sin^{-1}(0.7) &< \sqrt{2}(\frac{1}{\sqrt{2}} - 0.7) \\ &= 1 - 0.7\sqrt{2} \end{aligned}$$

$$\Rightarrow \sin^{-1}(0.7) > \frac{\pi}{4} - 1 + 0.7\sqrt{2}$$

**Example 3.17.** What is  $\sin(1)$ ?

Let  $f(x) = \sin x$ ,  $a = 1$ ,  $b = \frac{\pi}{3}$ . Then by the MVT:

$$\frac{\sin(\pi/3) - \sin(1)}{\pi/3 - 1} = \cos \xi < 1$$

$$\Rightarrow -(\frac{\pi}{3} - 1) < \frac{\sqrt{3}}{2} - \sin(1) < \frac{\pi}{3} - 1$$

$$\Rightarrow \frac{\sqrt{3}}{2} - \frac{\pi}{3} + 1 < \sin(1) < \frac{\sqrt{3}}{2} + \frac{\pi}{3} - 1$$

### 3.5 Taylor's Theorem

Can we generalise the mean value theorem for functions with many derivative?

Lecture 17

We can write

$$f(x) = f(a) + (x-a)f'(a) + \frac{1}{2}(x-a)^2 f''(a) + \cdots + \frac{1}{n!}(x-a)^n f^{(n)}(a) + E_n$$

Where  $E_n$  is the error after  $n+1$  terms of the Taylor series.

#### Theorem 3.18: Taylor's Theorem

The error term in the Taylor Expansion

$$E_n = \int_a^x \frac{f^{(n+1)}(t)(x-t)^n}{n!} dt = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi)$$

*Proof.* We do this by induction. Note that when  $n=0$  the RHS is

$$f(a) + \int_a^x f'(t) dt = f(a) + [f(t)]_a^x = f(a) + f(x) - f(a) = f(x)$$

and equals the LHS.

We assume that Taylor's theorem holds when  $n=k$ . Then integrating by parts, we have:

$$\begin{aligned} f(x) &= f(a) + (x-a)f'(a) + \cdots + \frac{1}{k!}(x-a)^k f^{(k)}(a) + \int_a^x \frac{f^{(k+1)}(t)(x-t)^k}{k!} dt \\ &= f(a) + \cdots + \left[ \frac{f^{(k+1)}(t)(x-t)^{k+1}}{k!(k+1)(-1)} \right]_a^x + \int_a^x \frac{f^{(k+2)}(t)(x-t)^{k+1}}{(k+1)!} dt \\ &= f(a) + \cdots + \frac{f^{(k+1)}(a)(x-a)^{k+1}}{k!(k+1)(-1)} + \int_a^x \frac{f^{(k+2)}(t)(x-t)^{k+1}}{(k+1)!} dt \end{aligned}$$

So the theorem also holds for  $n=k+1$ . Hence by induction it holds for all  $n \in \mathbb{N}$ .

Now by the Intermediate Value Theorem, in the interval  $[a, x]$ ,  $f^{(n+1)}$  attains a maximum value  $M$  and minimum value  $m$ :

$$m \leq f^{(n+1)}(t) \leq M$$

$$\implies \frac{m(x-t)^n}{n!} \leq \frac{f^{(n+1)}(t)(x-t)^n}{n!} \leq \frac{M(x-t)^n}{n!}$$

$$\implies \int_a^x \frac{m(x-t)^n}{n!} \leq \int_a^x \frac{f^{(n+1)}(t)(x-t)^n}{n!} \leq \int_a^x \frac{M(x-t)^n}{n!}$$

$$\Rightarrow \frac{m(x-a)^{n+1}}{(n+1)!} \leq \int_a^x \frac{f^{(n+1)}(t)(x-t)^n}{n!} \leq \frac{M(x-a)^{n+1}}{(n+1)!}$$

So for  $\xi \in (a, x)$ , we also have

$$E_n = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi) \quad \blacksquare$$

Note: If  $n = 0$ , Taylor's theorem becomes the Mean Value Theorem

$$f(x) = f(a) + E_0 = f(a) + \frac{(x-a)}{1!} f'(\xi)$$

**Example 3.19.** What is  $\sinh(1)$ ?

$$\sinh = x + \frac{x^3}{3!} + E$$

By Taylor's theorem:

$$E = \frac{x^5}{5!} (\sinh \xi)^{(5)} = \frac{x^5}{5!} \cosh \xi, \quad 0 < \xi < x$$

When  $x = 1$

$$E = \frac{1}{5!} \cosh \xi, \quad 0 < \xi < x$$

As  $\cosh \xi = \frac{e^\xi + e^{-\xi}}{2}$  is an increasing function, so

$$\cosh \xi < \frac{e + e^{-1}}{2} < e < 3$$

So

$$E < \frac{3}{5!} = \frac{3}{120} = \frac{1}{40} = 0.025$$

**Example 3.20.** Improve yesterday's approximation of  $\sin(1)$ .

Use Taylor series about  $x = \frac{\pi}{3} = a$  in formula:

$$\begin{aligned} \sin(1) &= \sin(\pi/3) + (1 - \pi/3) \cos(\pi/3) + \frac{1}{2}(1 - \pi/3)^2 (-\sin(\pi/3)) \\ &\quad + \frac{1}{6}(1 - \pi/3)^3 (-\cos \pi/3) + \frac{1}{4!}(1 - \pi/3)^4 \sin \xi \end{aligned}$$

where  $1 < \xi < \pi/3$ . So approximately:

$$\sin(1) \approx \frac{\sqrt{3}}{2} + \frac{1}{2}(1 - \pi/3) + \frac{1}{2}(1 - \pi/3)^2 \left(-\frac{\sqrt{3}}{2}\right) + \frac{1}{6}(1 - \pi/3)^3 - \frac{1}{2}$$

| Error  $< \frac{1}{24}(\frac{1}{20})^4$  since  $\frac{\pi}{3} \equiv 1.0472 < 1.05$ .

So we can reliably and methodically approximate any differentiable function near any point by a polynomial and guarantee that our error is as small as we like.

Note: General power series expansions about  $x = a$  take the form

$$f(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \dots$$

Let  $y = x - a$

$$f(y + a) = g(y) = c_0 + c_1y + c_2y^2 + \dots$$

If you are asked for an expansion about  $x = 1$ , say,

$$f(x) = c_0 + c_1(x - 1) + c_2(x - 1)^2 + \mathcal{O}(x - 1)^3$$

*Don't write this as  $d_0 + d_1x + d_2x^2 + \mathcal{O}(x - 1)^3$ ; this will lead to confusion over what you're expanding in (which may mislead you to break the golden rule.)*

## 3.6 Stationary Points

Most important practical use of differentiation is in finding maxima and minima of functions. Formally

Lecture 18

**Definition.** A *stationary point* of a differentiable function  $f(x)$  is a point  $(a, f'(a))$  where  $f'(a) = 0$ . Such a point may be maximum, minimum or neither.

Near such a point, we can expand (if  $f$  is suitably differentiable):

$$f(x) = f(a) + (x - a)f'(a) + \frac{1}{2}(x - a)^2 f''(a) + \frac{1}{6}(x - a)^3 f'''(a) + \dots$$

So at a stationary point

$$f(x) = f(a) + \frac{1}{2}(x - a)^2 f''(a) + \mathcal{O}(x - a)^3$$

Hence if

- $f''(a) > 0$  we have a local minimum.
- $f''(a) < 0$  we have a local maximum.
- $f''(a) = 0$  then we have...

$$f(x) = f(a) + \frac{1}{6}(x - a)^3 f'''(a) + \mathcal{O}(x - a)^4$$

- If  $f'''(a) \neq 0$ , then  $f(x)$  increases / decreases as we increase / decrease from  $x = a$  - this is neither a maximum or minimum.

– If  $f'(a) = 0 = f''(a) = f'''(a)$ , then we have

$$f(x) = f(a) + \frac{1}{24} f''''(a)(x-a)^4$$

Usually if  $f'(a) = 0$ , we look at the sign of  $f''(a)$  and that would be enough.

**Example 3.21.** Suppose

$$f'(x) = \sin x e^{-[(x^2+a^2)^{1/2} + \log[\sinh^{-1}(2x^3)]]}$$

Note that  $f'(\pi) = 0$ . What is  $f''(\pi)$ ?

At a point where  $u = 0$ ,  $(uv)' = u'(a)v(a) + 0$ , so

$$f''(\pi) = \cos \pi e^{-[(\pi^2+a^2)^{1/2} + \dots]}$$

Alternatively look at the sign of  $f'(a - \epsilon)$  and  $f'(a + \epsilon)$  as  $\epsilon > 0$ .

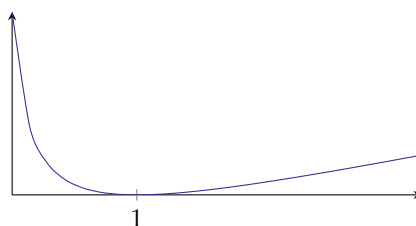
We can also use the fact that a differentiable function on a closed interval attains its maximum and minimum values *either* at an end point *or* at a stationary point.

**Example 3.22.**

$$f(x) = x + \frac{1}{x}$$

$$f'(a) = 1 - \frac{1}{x^2} = 0 \text{ at } x = \pm 1$$

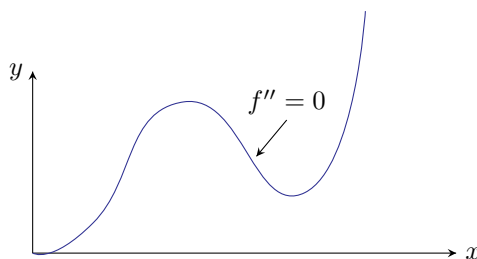
Looking at the graph...



So  $x = 1$  *must* be a minimum.

## Points of Inflexion / Inflection

**Definition.** A point of *inflexion* is where  $f''(x) = 0$ . It is *not* necessary for  $f'(x)$  to also be zero.



By Rolle's Theorem  $\exists$  a point where  $(f')' = 0$  between the zeros of  $f'$ .

### 3.7 Curve Plotting

It's very important to be able to give a schematic sketch of a function, i.e. the graph  $y = f(x)$ . This is *not* the best way of defining a curve.  $y = f(x)$  has only one  $y$  value for each  $x$  value.

#### Parametric Curves

**Definition.** The parametric definition is of the form

$$x = f(t), y = g(t) \} \text{ in } a \leq t \leq b$$

which defines *any* curve for suitable  $f, g$  and parameter  $t$ .

#### Example 3.23.

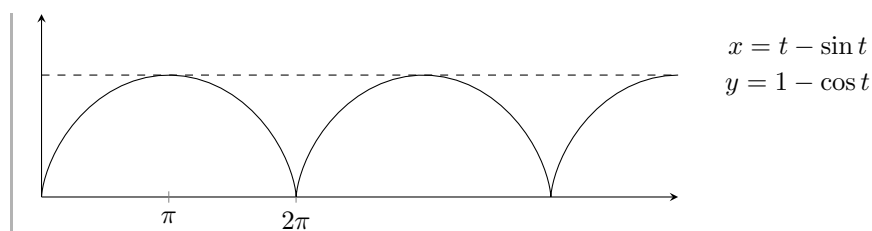
$$\left. \begin{aligned} x &= t - \sin t \\ y &= 1 - \cos t \end{aligned} \right\} 0 < t$$

What is  $\frac{dy}{dx}$ ?

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} / \frac{dx}{dt} \\ &= \frac{\sin t}{1 - \cos t} \\ &= \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \sin^2 \frac{t}{2}} \\ &= \cot \frac{t}{2} \end{aligned}$$

Now  $\frac{dy}{dx} = \cot \frac{t}{2}$  is infinite at  $t = 0$ , 0 at  $t = \pi$ , infinite at  $t = 2\pi$ .

At  $t = 0, x = 0, y = 0$ . Note that  $0 \leq y \leq 2$ .

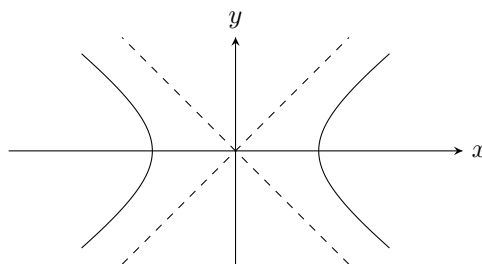


This plots the path of a point on the circumference of a wheel as it moves down a road - a *cycloid*.

There are more than one way of parameterising a curve.

Lecture 19

**Example 3.24** (Hyperbola). The rectangular hyperbola is  $x^2 - y^2 = 1$ .

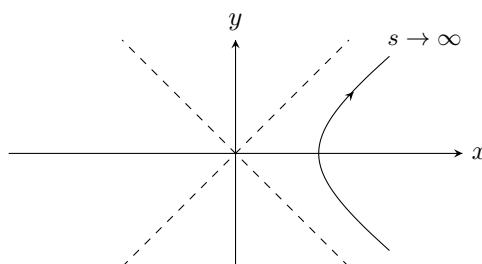


If we want to parametrise it, we could write

$$\left. \begin{aligned} x &= \cosh s \\ y &= \sinh s \end{aligned} \right\} \cosh^2 s - \sinh^2 s = 1$$

$$\implies x^2 - y^2 = 1$$

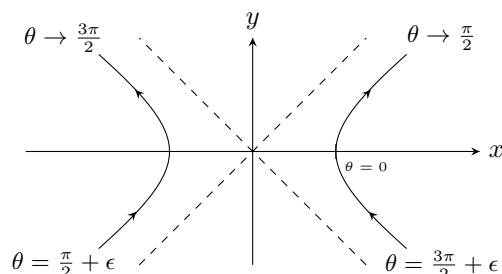
As  $\cosh s > 0$ , we now get  $x < 0$  using this parametrisation:



Instead we use  $\sec^2 \theta = 1 + \tan^2 \theta$ , i.e. write

$$\left. \begin{aligned} x &= \sec \theta \\ y &= \tan \theta \end{aligned} \right\} 0 \leq \theta < 2\pi$$

This parametrisation captures the whole curve.



There are others, such as  $x = \sec \alpha^3$ ,  $y = \tan \alpha^3$  for  $0 \leq \alpha \leq (2\pi)^{\frac{1}{3}}$ , or  $\sec(\beta + 3000)$ ,  $\tan(\beta + 3000)$ .

If  $x = f(\theta)$ ,  $y = g(\theta)$ . We can find  $\frac{dy}{dx} = \frac{g'}{f'}$  or

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left( \frac{g'}{f'} \right) \\ &= \frac{d\theta}{dx} \frac{d}{d\theta} \left( \frac{g'}{f'} \right) \\ &= \frac{1}{f'} \frac{f'g'' - g'f''}{f'^2} \end{aligned}$$

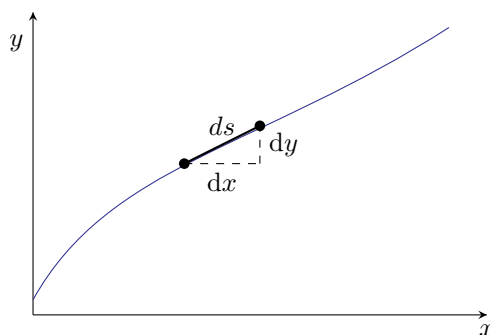
The second derivative  $y''$  is closely related to the *curvature* of a curve.

## Arc length

If I move along a curve  $y = y(x)$ , a small distance

$$(x, y) \rightarrow (x + \delta x, y + \delta y)$$

then the distance moved is  $\delta s$  where  $\delta s^2 = \delta x^2 + \delta y^2$ :



More formally:

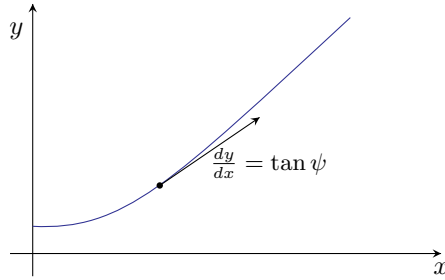
$$ds^2 = dx^2 + dy^2 \text{ is the limit as } \delta x \rightarrow 0$$

or

$$\left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

$$\lim_{\delta t \rightarrow 0} \left(\frac{\delta x}{\delta t}\right) = \frac{dx}{dt}$$

by definition.



$$\tan \psi = \frac{dy}{dx}, \cos \psi = \frac{dx}{ds}, \sin \psi = \frac{dy}{ds}$$

**Definition.**

$\left. \begin{array}{l} s \text{ arc length} \\ \psi \text{ tangent angle} \end{array} \right\}$  these are known as *intrinsic coordinates*

**Definition.** As we move along a curve,, the tangent angle,  $\psi$  may change. We define the *curvature*,  $\kappa$ , to be

$$\kappa = \frac{d\psi}{ds}$$

The rate of change of angle with distance along the curve.

*How can we relate this to Cartesian coordinates?*

$$\tan \psi = \frac{dy}{dx}, \cos \psi = \frac{dx}{ds}, \kappa = \frac{d\psi}{ds} = \frac{d\psi}{dx} \frac{dx}{ds}$$

Differentiate  $\tan \psi$  with respect to  $x$  implicitly

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx}(\tan \psi) = \sec^2 \psi \frac{d\psi}{dx}$$

Hence

$$\kappa = \frac{\cos \psi \frac{d^2 y}{dx^2}}{\sec^2 \psi} = \frac{y''}{\sec^2 \psi}$$

Now  $\sec^2 \psi = 1 + \tan^2 \psi = 1 + \left(\frac{dy}{dx}\right)^2$ . Hence

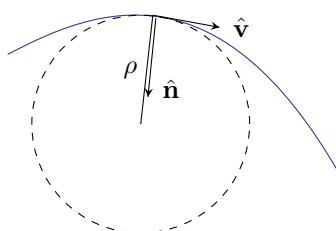
$$\boxed{\kappa = \frac{y''}{(1 + (y')^2)^{3/2}}}$$

where  $y' = \frac{dy}{dx}$ ,  $y'' = \frac{d^2y}{dx^2}$ .

**Definition.** We also define the *radius of curvature*.

$$\rho = \frac{1}{\kappa} = \frac{ds}{d\psi} = \frac{(1 + (y')^2)^{3/2}}{y''}$$

This is the radius of the osculating circle:



## Polar Coordinates

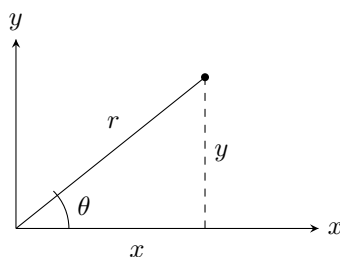
**Definition.** Given a point  $(x, y)$  we can define

Lecture 20

$$r = \sqrt{x^2 + y^2} \geq 0$$

and

$$\theta = \begin{cases} \arctan(y/x) & \text{if } x > 0 \\ \pi - \arctan(y/x) & \text{if } x < 0 \end{cases}$$



Our definition of  $\theta$  is cumbersome, ugly, confusing and messy, so we instead define

$$\theta := \begin{cases} \cos \theta & = \frac{x}{r} \\ \sin \theta & = \frac{y}{r} \end{cases} \quad \text{say } 0 \leq \theta < 2\pi$$

$$\implies x = r \cos \theta, y = r \sin \theta$$

We can define a curve by  $r = f(\theta)$  if we want.

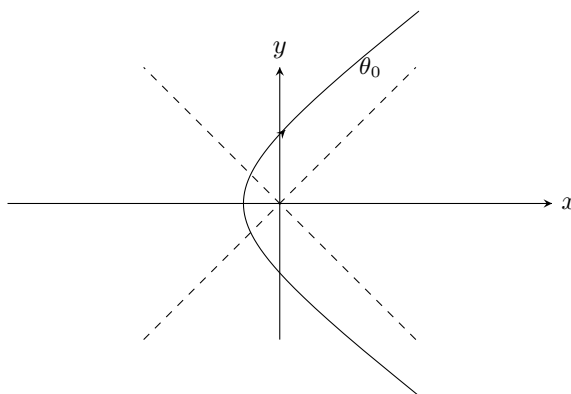
**Example 3.25.**

$$r = \frac{l}{1 + e \cos \theta}$$

Where  $l$  and  $e$  are constants. (See M1A1 MECHANICS)

*How do we plot such a curve?*

- (a) Join the dots. Find points on the curve to get an idea of how it looks. For  $r = \frac{1}{1 + e \cos \theta}$ , note that  $e \geq 1$ .  $1 + e \cos \theta$  may be zero.



- (b) Regard Polar Coordinates as a parametric definition.

We know

$$\begin{aligned} x &= r \cos \theta, y = r \sin \theta, r = \frac{1}{1 - 2 \cos \theta} \\ \implies x &= \frac{\cos \theta}{1 - 2 \cos \theta}, y = \frac{\sin \theta}{1 - 2 \cos \theta} \end{aligned}$$

- (c) Try to transform to Cartesian.

$$\begin{aligned} r &= \frac{1}{1 - 2 \cos \theta} \implies r - 2r \cos \theta = 1 \\ \implies r &= 1 + 2x \\ \iff \sqrt{x^2 + y^2} &= 1 + 2x \\ \implies x^2 + y^2 &= (1 + 2x)^2 = 4x^2 + 4x + 1 \\ \implies 3x^2 + 2x - y^2 + 1 &= 0 \end{aligned}$$

Squaring introduces spurious branch of hyperbola. We need to try another approach...

$$\begin{aligned}
 r = \frac{1}{1 + e \cos \theta} &\implies r + ex = 1 \\
 &\implies r^2 = (1 - ex)^2 \\
 &\implies x^2 + y^2 = 1 - 2ex + e^2 x^2 \\
 &\implies x^2(1 - e^2) + 2ex + y^2 = 1
 \end{aligned}$$

We have a conic:

- $e = 0$  - Circle
- $e = 1$  - Parabola
- $0 < e < 1$  - Ellipse
- $e > 1$  - Hyperbola

$$\begin{aligned}
 x^2 + \frac{2e}{1 - e^2}x + \frac{y^2}{1 - e^2} &= \frac{1}{1 - e^2} \\
 \implies \left(x + \frac{e}{1 - e^2}\right)^2 + \frac{y^2}{1 - e^2} &= \frac{1}{1 - e^2}
 \end{aligned}$$

So the foci is at  $\left(-\frac{e}{1 - e^2}, 0\right)$ .

Sometimes we cannot easily transform to Cartesian. e.g.  $r = \frac{1}{\theta}$  for  $\theta > 0$ .

## 4 Integration

### 4.1 The Riemann Integral

*Note: Covered rigorously in M2PM2 ANALYSIS II.*

**Definition.** Given an interval  $[a, b]$ , we define a *partition* to be a set of  $n$  points,  $x_1, x_2, \dots, x_n$  such that

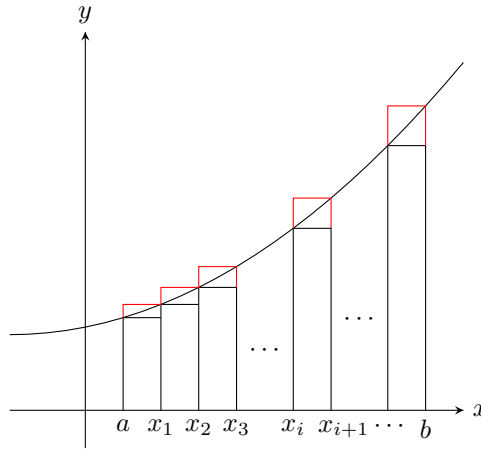
Lecture 21

$$a \equiv x_0 < x_1 < x_2 < \dots < x_n < x_{n+1} \equiv b$$

For a given partition, we choose points on each subinterval,  $\xi_0, \xi_1, \dots, \xi_n$  such that for all  $i$ ,  $x_i < \xi_i < x_{i+1}$ . Then for each function  $f(x)$ , we define the *Riemann sum* to be

$$S_n = (x_1 - x_0)f(\xi_0) + (x_2 - x_1)f(\xi_1) + \dots + (x_{n+1} - x_n)f(\xi_n)$$

Pictorially, we are forming  $n+1$  rectangles whose sum resembles the area under the curve  $y = f(x)$ . The upper sum is the total area of the red rectangles, while the lower sum is the total area of the black rectangles:



We now let  $n \rightarrow \infty$  in such a manner than  $(x_{i+1} - x_i) \rightarrow 0$  for all  $i$ .

**Definition.** If the sequence  $S_n$  tends to a limit, and if this limit does not depend on the particular partitions nor the value of  $\xi$  we choose, we can write this as the *definite integral* of  $f(x)$  between  $x = a$  and  $x = b$ .

$$\lim_{n \rightarrow \infty} S_n = \int_a^b f(x) \, dx$$

The function  $f(x)$  is the *integrand*.

As the integral is a generalisation of a sum, it behaves similarly to one. Various properties follow from the definition, for example:

**Theorem 4.1: Mean Value Theorem for Integrals**

Suppose  $f$  is integrable in  $[a, b]$ . Then  $\exists \xi \in (a, b)$ , such that

$$\int_a^b f(x) \, dx = (b - a)f(\xi)$$

*Proof.* Being integrable,  $f$  is bounded by  $m \leq f(x) \leq M$ , so then

$$m(b - a) \leq \int_a^b f(x) \, dx \leq M(b - a)$$

Suppose  $m$  and  $M$  are the minimum and maximum values attained by a continuous function  $f$  over  $[a, b]$  then  $(b - a)f(x)$  attains every value between  $(b - a)m$  and  $(b - a)M$  in  $[a, b]$ . In particular the value equal to the integral. ■

**Theorem 4.2: Fundamental Theorem of Calculus**

Differentiation is the reverse of integration:

$$\frac{d}{dx} \int_a^x f(t) \, dt = f(x)$$

*Proof.* If we fix the lower limit  $a$  then  $f(x)$  defines another function:

$$F(x) = \int_a^x f(t) \, dt$$

It follows that for any  $c$  and  $d$

$$\int_c^d f(t) \, dt = \int_a^d f(t) \, dt - \int_a^c f(t) \, dt = F(d) - F(c).$$

Consider now

$$\int_x^{x+h} f(t) \, dt = F(x + h) - F(x)$$

By the Mean Value Theorem for Integrals (Theorem 4.2),  $\exists \xi \in (x, x + h)$  such that

$$F(x + h) - F(x) = (x + h - x)f(\xi)$$

Thus

$$\lim_{h \rightarrow 0} \left[ \frac{F(x + h) - F(x)}{h} \right] = \lim_{h \rightarrow 0} f(\xi)$$

As  $h \rightarrow 0$ ,  $\xi \rightarrow x$ . Thus the limit on the RHS exists and equals  $f(x)$ , and so  $F(x)$  is differentiable with derivative  $f(x)$ , so

$$\frac{d}{dx} \int_a^x f(t) \, dt = f(x) \quad \blacksquare$$

What sort of functions are integrable?

- (a) Continuous functions.  $f$  continuous on  $[a, b] \implies \int_a^b f(x) dx$  exists
  - (b) Functions with a single *finite* jump
  - (c) Functions with a finite number of finite jumps
  - (d) If there are an infinite number of discontinuities we're not sure. Integrable.
  - (e) If the function has an infinite discontinuity (i.e. it is unbounded) integral may or may not exist.
  - (f) If either or both limit ( $a$  or  $b$ ) is infinite, the integral may or may not exist.
- (e) and (f) are important.

## 4.2 Integrals over infinite ranges

Does  $\int_0^\infty f(x) dx$  exist?

Assume  $f(x)$  is continuous.

$$\int_0^\infty f(x) dx = \lim_{N \rightarrow \infty} \int_0^N f(x) dx$$

Clearly

- (a)  $f(x) = x$ , the limit does not exist  $\int_0^N x dx = \frac{1}{2}N^2 \rightarrow \infty$
- (b)  $f(x) = 1$  limit does not exist similarly.
- (c)  $f(x) = \sin x$

We might surmise that a necessary condition for  $\int_0^\infty f(x) dx$  to exist is that  $f(x) \rightarrow 0$  as  $x \rightarrow \infty$ . In fact this is not true.  $\int_0^\infty \sin(x^2) dx$  does not exist - this is not obvious.

What is true, is that if  $\lim_{x \rightarrow \infty} f(x) = A \neq 0$ , then integral does not exist.

Suppose  $f(x) \rightarrow 0$  as  $x \rightarrow \infty$ . In particular, if  $f(x) \approx \frac{1}{x^\alpha}$  as  $x \rightarrow \infty$ , then  $\int_0^\infty f(x) dx$  exists iff  $\alpha > 1$ .

### Theorem 4.3

If  $f(x)$  is integrable and  $|f(x)| < \frac{A}{x^\alpha}$  as  $x \rightarrow \infty$  if  $\alpha > 1$ . Then  $\int_0^\infty f(x) dx$  exists.

Lecture 22

What if  $f(x)$  has an infinite discontinuity?

E.g.

$$f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

Does  $\int_{-1}^1 f(x) dx$  exist?

Look at

$$\int_{-1}^{-\delta} + \int_{-\delta}^{\epsilon} + \int_{\epsilon}^1, \text{ where } \delta, \epsilon > 0.$$

So only the behaviour very close to the singularity is important.

Consider

$$\int_0^1 \frac{1}{x^\alpha} dx, \text{ define as } \lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^1 \frac{1}{x^\alpha} dx$$

Then we consider

$$\frac{[x^{1-\alpha}]_{\epsilon}^1}{1-\alpha} = \frac{1 - \epsilon^{1-\alpha}}{1-\alpha}$$

Exercise: Calculate  $\lim_{\alpha \rightarrow 1}$  of the equation.

Now let  $\epsilon \rightarrow 0$ :

- If  $\alpha > 0$ ,  $\epsilon^{1-\alpha} \rightarrow \infty \implies$  integral does not exist.
- If  $\alpha < 1$ ,  $\epsilon^{1-\alpha} \rightarrow 0 \implies$  integral does exist.
- If  $\alpha = 1$ , we get  $\int_{\epsilon}^1 \frac{1}{x} dx = \log \epsilon \rightarrow \infty$  as  $\epsilon \rightarrow 0$ .

So if  $f \rightarrow \infty$  at  $x = 0$  and otherwise continuous,  $\int_0^1 f dx$  will exist if  $f \approx \frac{1}{x^\alpha}$  where  $\alpha < 1$  near  $x = 0$ . Similarly if  $f(x)$  has a singularity at  $x = c$  say, we need  $f \approx \frac{1}{(x-c)^\alpha}$  where  $\alpha < 1$  near  $x = c$  for integral to exist.

**Example 4.4.**

$$\int_0^x \frac{e^x}{(x-1)^{1/3}} dx$$

The only problem is at  $x = 1$ . Near  $x = 1$ ,

$$\text{Integral} \approx \int \frac{e}{(x-1)^{1/3}} dx$$

The power is  $-\frac{1}{3}$  at the singularity, so as  $\alpha = \frac{1}{3} < 1 \implies$  integral does exist.

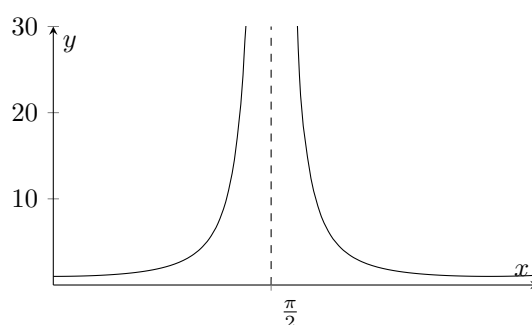
Functions with a finite number of (“ $\alpha < 1$ ”), “not too bad” infinite singularities as integrable.

**Example 4.5.**

$$\begin{aligned}
 \int_0^{\frac{3\pi}{4}} \sec^2 x \, dx &= [\tan x]_0^{\frac{3\pi}{4}} \\
 &= \tan \frac{3\pi}{4} - \tan 0 \\
 &= -1
 \end{aligned}$$

But...  $\sec^2 x > 0 \implies \int \sec^2 x > 0$ .

So what has gone wrong?  $\sec^2 x$  has a singularity at  $x = \frac{\pi}{2}$ :



vc

$$\begin{aligned}
 \cos x &\approx \cos \frac{\pi}{2} + (x - \frac{\pi}{2})(-1) + \frac{1}{2}(x - \frac{\pi}{2})^2 \cdot 0 \\
 &= 0 + (\frac{\pi}{2} - x)
 \end{aligned}$$

Near  $x = \frac{\pi}{2}$ ,  $\sec^2 x = \frac{1}{(\frac{\pi}{2} - x)^2}$ . So  $\alpha = 2 > 1 \implies$  not integrable.

Hence

$$\int_0^{\frac{3\pi}{4}} \sec^2 x \, dx = \text{"}\infty\text{"} \neq -1$$



*Warning.* Important to look at Infinities!!

#### Theorem 4.6: Integration by Substitution

$$\int_{\phi(a)}^{\phi(b)} f(x) \, dx = \int_a^b f(\phi(t)) \phi'(t) \, dt$$

*Proof.* Using the chain rule:

$$\begin{aligned}
 \frac{d}{dx} [F(\phi(t))] &= F'(\phi(t)) \cdot \phi'(t) \\
 &= f(\phi(t)) \cdot \phi'(t)
 \end{aligned}$$

Suppose  $F(x) = \int^x f(s) \, ds$ , then  $F'(x) = f(x)$ . Now integrating with respect

to  $t$ :

$$\begin{aligned}\int_a^b \frac{d}{dt} [F(\phi(t))] dt &= \int_a^b f(\phi(t)) \cdot \phi'(t) dt \\ F(\phi(b)) - F(\phi(a)) &= \int_a^b f(\phi(t)) \cdot \phi'(t) dt \\ \int_{\phi(a)}^{\phi(b)} f(x) dx &+ \int_a^b f(\phi(t)) \cdot \phi'(t) dt \quad \blacksquare\end{aligned}$$

So starting with

$$\int_{\phi(a)}^{\phi(b)} f(x) dx$$

we can make a substitution  $x = \phi(t)$ . Replace  $dx$  by  $\phi'(t) dt$  and replace the limits accordingly. This is very useful for evaluating integrals. But this is even more useful:

**Theorem 4.7: Integration by Parts**

$$\int_a^b uv' dx = [uv]_a^b - \int_a^b u'v dx$$

*Proof.* Recall the product rule:  $\frac{d}{dx}(uv) = \frac{du}{dx}v + \frac{dv}{dx}u$ .

$$\begin{aligned}\int_a^b \frac{d}{dx}(uv) dx &= \int_a^b \frac{du}{dx}v dx + \int_a^b u \frac{dv}{dx} dx \\ [uv]_a^b &= \int_a^b u'v dx + \int_a^b uv' dx \\ \Rightarrow \int_a^b uv' dx &= [uv]_a^b - \int_a^b u'v dx \quad \blacksquare\end{aligned}$$

This enables us to transform an integral into another integral which may be easier to evaluate, for example:

**Example 4.8.**

$$\int_0^1 \tan^{-1} x dx = I$$

Lecture 23

Treat this as  $1 \cdot \tan^{-1} x - \tan^{-1} x$  is easy to differentiate.

$$\begin{aligned} I &= [x \tan^{-1} x]_0^1 - \int_0^1 x \frac{1}{1+x^2} dx \\ &= \pi - 0 - \left[ \frac{1}{2} \log(1+x^2) \right]_0^1 \\ &= \frac{\pi}{4} - \frac{1}{2} \log 2 \end{aligned}$$

### 4.3 Evaluation of Integrals

Any integral is an answer - it defines a function in its own right. *Some* (only a few) integrals can be expressed in terms of known functions, e.g.  $\sin$ ,  $\log$ ,  $\sqrt{\quad}$  etc. Such simplifications are useful - we call this *evaluating* the integral.

*What kind of integrals can we evaluate?*

- (a) Polynomials  $\rightarrow$  Polynomials
- (b) Any power series  $\rightarrow$  Another power series
- (c) Rational functions  $\frac{P(x)}{Q(x)}$ , where  $P, Q$  are Polynomials.

We use partial fractions to express rational fractions as

$$\sum_{i=1}^N \frac{A_i}{x - \alpha_i} + \frac{B_i x + c_i}{x^2 + \beta_i + \gamma_i} \text{ etc.}$$

**Example 4.9.**

$$\begin{aligned} \int \frac{dx}{x^3 + 1} &= \int \frac{dx}{(x+1)(x^2 - x + 1)} \\ &= \int \frac{1/3}{x+1} + \frac{-1/3x + 2/3}{x^2 - x + 1} dx \end{aligned}$$

### 4.4 Reduction Formulae

If we were an integral with a parameter,  $n$  (usually an integer), we can sometimes relate it to a similar integral with a smaller value of  $a$  (usually by integrating by parts)

Lecture 24

**Example 4.10.**

$$\begin{aligned}
I_n &= \int_0^{\frac{\pi}{2}} \sin^n x \, dx \quad n \geq 1 \\
&= \int_1^{\frac{\pi}{2}} \sin x \sin^{n-1} x \, dx \\
&= [-\cos x - \sin^{-1} x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x)(n-1) \sin^{n-2} x \cdot \cos x \, dx \\
&= \int_0^{\frac{\pi}{2}} (n-1) \cos^2 x \sin^{n-2} x \, dx \\
&= \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \sin^{n-2} x \, dx \\
&= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x \, dx \\
&= (n-1) I_{n-2} - (n-1) I_n \\
\implies I_n &= \frac{(n-1)}{n} I_{n-2}
\end{aligned}$$

So we end up with a reduction formula. This can be applied repeatedly, i.e.

$$I_n = \left( \frac{n-1}{n} \right) I_{n-2} = \left( \frac{n-1}{n} \right) \left( \frac{n-3}{n-2} \right) I_{n-4}$$

So we can say

$$I_n = \left( \frac{n-1}{n} \right) \left( \frac{n-3}{n-2} \right) \left( \frac{n-5}{n-4} \right) \cdots \begin{cases} \frac{4}{5} \cdot \frac{2}{3} I_1 & \text{if } n \text{ is odd} \\ \frac{4}{4} \cdot \frac{1}{2} I_0 & \text{if } n \text{ is even} \end{cases}$$

Now

$$\begin{aligned}
I_1 &= \int_0^{\frac{\pi}{2}} (\sin x)^0 \, dx = \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2} \\
I_0 &= \int_0^{\frac{\pi}{2}} \sin x \, dx = -\cos x \Big|_0^{\frac{\pi}{2}} = 0 - (-1) = 1
\end{aligned}$$

e.g.

$$\int_0^{\frac{\pi}{2}} \sin^{10} x \, dx = \frac{9}{10} \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

and

$$\int_0^{\frac{\pi}{2}} \sin^9 x \, dx = \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

What about  $\int_0^{\frac{\pi}{2}} \cos^{10} x \, dx$ ? Let  $y = \frac{\pi}{2} - x$ , then  $I_n = \int_0^{\frac{\pi}{2}} \sin^{10} y \, dy$ , yielding the same formula.

There are many others, techniques are similar.

**Example 4.11.**

$$\int_0^{\frac{\pi}{4}} \sec^n x \, dx, \quad \int_0^1 x^m(1-x)^n \, dx \text{ etc.}$$

Consider

$$\int_0^\infty e^{-\alpha x} \, dx \quad (\alpha > 0) = \left[ \frac{e^{-\alpha x}}{-\alpha} \right]_0^\infty = \frac{1}{\alpha}$$

Then differentiating w.r.t.  $\alpha$

$$\begin{aligned} \frac{1}{\alpha} &= \int_0^\infty e^{-\alpha x} \, dx \\ \Rightarrow \frac{-1}{\alpha^2} &= \frac{d}{d\alpha} \int_0^\infty e^{-\alpha x} \, dx \\ &= \int_0^\infty -x e^{-\alpha x} \, dx \end{aligned}$$

Repeating this

$$\begin{aligned} \frac{2}{\alpha^3} &= \int_0^\infty (-x)^2 e^{-\alpha x} \, dx \\ &\vdots \\ (-1)^n \frac{n!}{\alpha^{n+1}} &= \int_0^\infty (-x)^n e^{-\alpha x} \, dx \end{aligned}$$

Now put  $\alpha = 1$ , we get

$$\boxed{n! = \int_0^\infty x^n e^{-x} \, dx}$$

Recall that the MVT for integrals is

$$\int_a^b f(x) \, dx = (b-a)f(\xi)$$

for some  $a < \xi < b$ . i.e.

$$\int_a^b f(\xi) \, dx = \frac{1}{b-a} \int_a^b f(x) \, dx$$

So  $\xi$  is the *average* of  $f(x)$  or the mean value. Hence the name of the theorem.

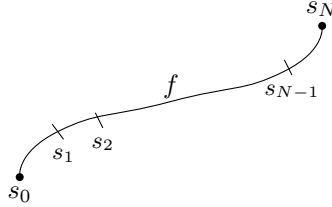
We can also define average values for functions defined on a line of any shape or an area or volume.

## 4.5 Path Integrals

Consider

$$\int_c f \, ds$$

where  $ds$  is the arc length and  $f$  is some function defined on the curve.



$$\int_0^c f \, ds = \sum f(s_n)(s_{n+1} - s_n)$$

as before, we can calculate the limit of the sum of small steps.

*How can we calculate it?*

$$\int_c f \, ds = \int_{x_0}^{x_1} f \frac{ds}{dx} \, dx$$

Now from a right angled triangle, we know that

$$\left(\frac{ds}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

we can then try to find

$$\int_{x_0}^{x_1} f \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx = \text{integral of } f \text{ along the path defined by } y(x)$$

Set  $f = 1$  and we get the *length* of any curve.

$$L = \int_0^1 f \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

**Example 4.12.** Find the length of the quarter circle  $x^2 + y^2 = 1$ .

$$L = \int_{x_0}^{x_1} f \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

Now for  $y = \sqrt{1 - x^2}$

$$\frac{dy}{dx} = \frac{-x}{\sqrt{1 - x^2}}$$

Substitute in and evaluate the integral. We expect to get  $\frac{\pi}{2}$ :

$$\begin{aligned} L &= \int_0^1 \sqrt{1 + \frac{x^2}{1-x^2}} dx \\ &= \int_0^1 \sqrt{\frac{1-x^2+x^2}{1-x^2}} dx \\ &= \int_0^1 \sqrt{\frac{1}{1-x^2}} dx \\ &= \arcsin|_0^1 = \frac{\pi}{2} \end{aligned}$$

The problem with evaluating  $\int x ds$  of a half circle is that we have  $\pm$  values of  $y$ ... It's better to define the curve parametrically. Lecture 25

*How do we do this?*

Say let  $x = \cos \theta$ ,  $y = \sin \theta$  for  $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ . In general

$$\begin{aligned} \frac{ds^2}{d\theta^2} &= \frac{dx^2}{d\theta^2} + \frac{dy^2}{d\theta^2} \\ \Rightarrow ds &= \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \end{aligned}$$

So

$$\begin{aligned} \int_c x ds &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta \sqrt{(\sin \theta)^2 + (\cos \theta)^2} d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta d\theta \\ &= \sin \theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2 \end{aligned}$$

This is in some sense the “average”  $x$  value over  $c$ . Divide by the length of the curve  $= \int_c 1 ds = \pi$ , so  $\frac{2}{\pi}$  is the mean  $x$  value of  $c$ . This is the  $x$ -coordinate of the *centre of mass* (See M1A1 MECHANICS for examples)

$$\begin{aligned} \int f ds &= \int f \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int f \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \end{aligned}$$

for any parameter  $t$ .

The generalisation to 3D is easy... Suppose  $x, y$  and  $z$  are given parametrically in terms of  $t$ . Then the length of the curve is described by:

$$\int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

**Example 4.13.** If  $x = y$ ,  $y = \cos t$ ,  $z = \sin t$  this a helix. The length of a helix is

$$\int_{t_0}^{t_1} \sqrt{1 + \sin^2 t + \cos^2 t} dt = \sqrt{2}(t_1 - t_0)$$

## 4.6 Area of Integrals

Suppose we have a rectangle in the  $x - y$  plane. Divide into many smaller rectangles and evaluate a function  $f(x, y)$  in each of the small rectangles and add up the answers.

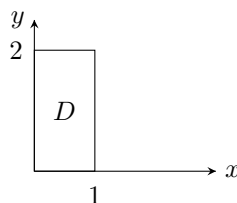
$$\sum_m \sum_n f$$

Take the limit as  $m, n \rightarrow \infty$ . if it works, we define an area integral:

$$\iint f dx dy = \iint f dA$$

*Can we evaluate this area integral easily?*

**Example 4.14.** Suppose  $D$  is the rectangle bounded by  $x = (1, 0)$ ,  $y = (0, 2)$ . What is  $\iint_D (y^2 = xy) dx dy$ ?



We argue that we can first treat  $x$  as a constant and integrate  $dy$  and then later integrate  $dx$ . We are just adding small rectangles in a particular way.

So keeping  $x$  constant and integrating  $dy$  first:

$$\begin{aligned} \iint f dx dy &= \int \left( \int f dy \right) dx \\ &= \int \left( \int f dx \right) dy \end{aligned}$$

$$\begin{aligned}
\Rightarrow \iint_D y^2 + xy \, dx \, dy &= \int_0^1 \left( \int_0^2 (y^2 + xy) \, dy \right) dx \\
&= \int_0^1 \left[ \frac{1}{3}y^3 + \frac{1}{2}xy^2 \right]_0^2 dx \\
&= \int_0^1 \left( \frac{8}{3} + 2x \right) dx \\
&= \frac{8}{3} - 2 \\
&= -\frac{11}{5}
\end{aligned}$$

What if we do it the other way round?

Then

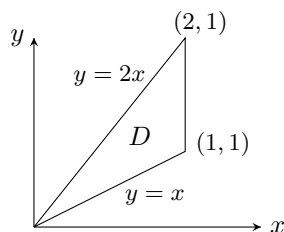
$$\iint_D (y^2 + xy) \, dx \, dy = \int_0^2 \left( \int_0^1 (y^2 + xy) \, dx \right) dy = -\frac{11}{5}$$

This works! *Mirabile dicta!* The fact that  $D$  was rectangular was very useful because the limits in each integral were constants. What about:

**Example 4.15.**

$$I = \iint_D x \, dA$$

Where  $A$  is the triangle bounded by  $(0,0)$ ,  $(1,1)$  and  $(2,1)$ , as in the picture:

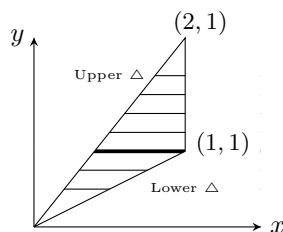


Taking horizontal strips is harder than vertical strips because of the change in limit as we'll see...

VERTICAL STRIPS. Keep  $x$  a constant and integrate  $dy$ :

$$\begin{aligned}
 I &= \int_0^1 \left( \int_x^{2x} x \, dy \right) dx \\
 &= \int_0^1 [xy]_x^{2x} dx \\
 &= \int_0^1 (x^2) dx \\
 &= \left[ \frac{1}{3} x^3 \right]_0^1 = \frac{1}{3}
 \end{aligned}$$

HORIZONTAL STRIPS. Keep  $y$  constant:

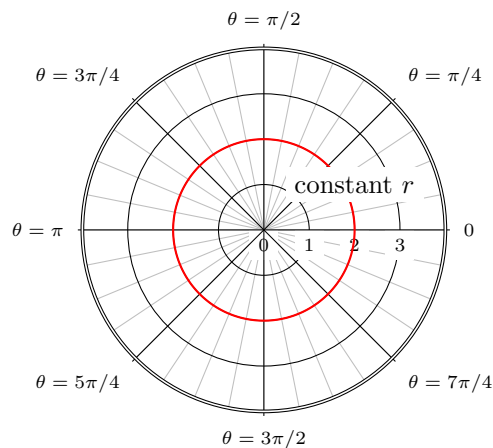


$$\begin{aligned}
 I &= \underbrace{\int_0^1 \left( \int_{\frac{y}{2}}^y x \, dx \right) dy}_{\text{Lower } \Delta} + \underbrace{\int_1^2 \left( \int_{\frac{y}{2}}^1 x \, dx \right) dy}_{\text{Upper } \Delta} \\
 &= \int_0^1 \left[ \frac{1}{2} x^2 \right]_{\frac{y}{2}}^y dy + \int_1^2 \left[ \frac{1}{2} x^2 \right]_{\frac{y}{2}}^1 dy \\
 &= \int_0^1 \frac{3}{8} y^2 dy + \int_1^2 \left( \frac{1}{2} - \frac{1}{8} y^2 \right) dy \\
 &= \left[ \frac{1}{8} y^3 \right]_0^1 + \left[ \frac{1}{2} y - \frac{1}{24} y^3 \right]_1^2 \\
 &= \frac{1}{8} + 1 - \frac{1}{3} - \frac{1}{2} + \frac{1}{24} \\
 &= \frac{8}{24} = \frac{1}{3}
 \end{aligned}$$

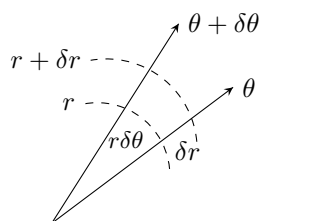
Changing variables in area integrals is possible but tricky - see M2AA2 next year. We can write  $dA$  in terms of any variables not just  $dx \, dy$ . The only case we'll consider is polar coordinates.

Lecture 26

## Polar coordinates



We argue that a small area element  $\delta A \approx \delta r \times (r\delta\theta)$  (almost a rectangle.)

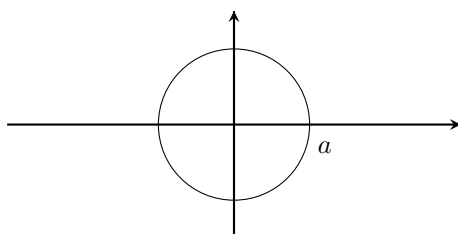


This suggests that  $dA = r dr d\theta$ . (as well as  $dA = dx dy$ ).

So (for suitable limits)

$$\iint_D f dA = \iint f r dr d\theta$$

Polar coordinates are useful if the domain of integration,  $D$ , has circular boundaries, e.g.  $D : r \leq a$ :



**Example 4.16.** Find

$$\int 1 dA$$

over the unit circle.

$$\begin{aligned}
 \int 1 \, dA &= \int_0^{2\pi} \int_0^a 1 \, r \, dr \, d\theta \\
 &= \int_0^{2\pi} \left( \int_0^a r \, dr \right) d\theta \\
 &= \int_0^{2\pi} \frac{1}{2} a^2 \, d\theta \\
 &= \frac{1}{2} a^2 [\theta]_0^{2\pi} \\
 &= \pi a^2
 \end{aligned}$$

**Example 4.17.** If  $f = x^2 + xy$ , what is  $\iint f \, dA$  over a circle?

Use polars.  $x = r \cos \theta$ ,  $y = r \sin \theta$ , so  $f = r^2(\cos^2 \theta + \sin \theta \cos \theta)$  and so

$$\begin{aligned}
 I &= \iint f \, dA = \iint r^2(\cos^2 \theta + \sin \theta \cos \theta) r \, dr \, d\theta \\
 &= \int_0^{2\pi} (\cos^2 \theta + \sin \theta \cos \theta) \left( \int_0^a r^3 \, dr \right) d\theta \\
 &= \int_0^{2\pi} (\cos^2 \theta + \sin \theta \cos \theta) \left( \frac{1}{4} a^4 \right) d\theta \\
 \implies I &= \frac{1}{4} a^4 \int_0^{2\pi} (\cos^2 \theta - \sin \theta \cos \theta) d\theta \\
 &= \frac{1}{4} a^4 \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} - \sin \theta \cos \theta d\theta \\
 &= \frac{1}{4} a^4 \left[ \frac{\theta}{2} + \frac{1}{4} \sin 2\theta + \frac{1}{2} \sin^2 \theta \right]_0^{2\pi} \\
 &= \frac{\pi}{4} a^4
 \end{aligned}$$

Note in the last example that that  $\frac{1}{2} \sin^2 \theta \Big|_0^{2\pi} = \int xy \, dA$  is 0. *Why?* We worked out the integral over a circle, which is an odd function of  $x$  and of  $y$ , so since the domain is symmetric, the term is 0.

Also note the integral of  $\cos^2$  over a whole period is  $\frac{1}{2}$ .

## 4.7 Volume Integrals

We can generalise to 3-D. E.g. If we want to integrate  $f(x, y, z)$  over the box  $0 < x < 1$ ,  $0 < y < 1$ ,  $0 < z < 1$ , we could write it as

$$\iiint f \, dV = \int_0^1 \int_0^1 \int_0^1 f \, dx \, dy \, dz$$

**Example 4.18.** E.g.  $f = xy$

$$\begin{aligned}
 \iiint f \, dV &= \int_0^1 \int_0^1 \int_0^1 xy \, dx \, dy \, dz \\
 &= \int_0^1 \int_0^1 \left[ \frac{1}{2} x^2 y \right]_0^1 dy \, dz \\
 &= \int_0^1 \int_0^1 \frac{1}{2} y \, dy \, dz \\
 &= \int_0^1 \left[ \frac{1}{4} y^2 \right]_0^1 dz \\
 &= \frac{1}{4}
 \end{aligned}$$

Note that

$$\iint dA = \iint 1 \, dy \, dx = \int [y] \, dx$$

So the area under a curve can be derived from a double integral. Likewise, an integral over an area can be interpreted as a value.

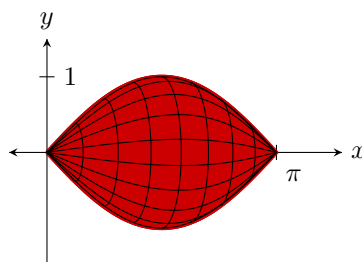
$$\iint f \, dA = \text{Volume between surface}$$

$z = f(x, y)$  and  $z = 0$ .

But volume integrals  $\longrightarrow$  4 dimensions is hard.

## Volumes of Revolution

Suppose the curve  $y = f(x)$  is rotated about the  $x$ -axis through  $2\pi$ . What is the volume formed?



We argue that a thin slice of thickness  $\delta x$  has a volume  $\pi y^2 \delta x$ , and hence the total volume is

$$\int_a^b \pi y^2 \, dx$$

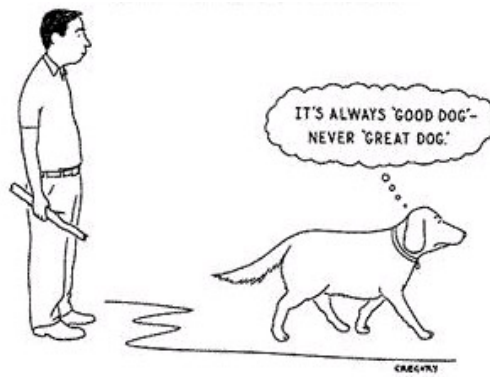
**Example 4.19.** Take a semicircle  $x^2 + y^2 = a^2$  for  $y \geq 0$ .

What is the volume when this is rotated about the  $x$ -axis?

$$\begin{aligned}
 V &= \int_{-a}^a \pi y^2 \, dx \\
 &= \int_{-a}^a \pi(a^2 - x^2) \, dx \\
 &= \pi \left[ a^2x - \frac{1}{3}x^3 \right]_{-a}^a \\
 &= \pi \left[ 2a^3 - \frac{2}{3}a^3 \right] \\
 &= \frac{4}{3}\pi a^3 \text{ (The volume of a sphere)}
 \end{aligned}$$

Likewise the surface of revolution can be shown to be

$$\int 2\pi y \, ds = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$



## 5 Complex Numbers

**Definition.**  $i = \sqrt{-1}$ .

Lecture 26

We call  $z$  a complex number iff  $z = x + iy$ , where  $x, y$  are real. Call  $x = \Re(z)$ ,  $y = \Im(z)$ . Note that  $\Im(z)$  is real (by definition).

The standard form:  $z = x + iy = r(\cos \theta + i \sin \theta)$ , the polar form.

Important result:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

You **must** be able to transform from standard form to polar form easily.

$$(x, y) = (r \cos \theta, r \sin \theta) \implies r^2 = x^2 + y^2, r = \sqrt{x^2 + y^2}.$$

**Definition.**  $r = |z|$  is the *modulus*,  $\theta = \text{Arg}(z)$ , the *argument*, say  $-\pi < \theta \leq \pi$ .

Complex numbers allow us to solve more equations.

**Example 5.1.** Solve  $z^6 - iz^3 - 1 = 0$ .

Write as  $(z^3)^6 - i(z^3) - 1 = 0$ . All the normal rules of algebra apply, but we also have  $i^2 = -1$ , so

$$z^3 = \frac{+i \pm \sqrt{(-i)^2 - 4(-1)}}{2} = \frac{i \pm \sqrt{3}}{2}$$

Now write the RHS  $\frac{i \pm \sqrt{3}}{2}$  in polar form.

$$\begin{aligned} re^{i\theta} &= \frac{i \pm \sqrt{3}}{2} \\ \implies r &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1 \end{aligned}$$

$$\left. \begin{aligned} \cos \theta &= \pm \sqrt{3}/2 \\ \sin \theta &= \frac{1}{2} \end{aligned} \right\} \implies \theta = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

So  $z^3 = e^{i\theta}$  where  $\theta = \frac{\pi}{6}$  or  $\frac{5\pi}{6} \implies z = e^{i\theta/3} = e^{i\pi/18}$  or  $e^{5\pi i/18}$  or other solutions:

Recall roots of unity (M1F),  $e^{2\pi} = 1$ . So  $z^3 = e^{i\theta} \cdot e^{2k\pi i}$  for integers  $k$ .  $z = e^{i(\theta+2k\pi)/3}$  gets different answers for  $k = 0, 1, 2$ , i.e. the answers are

$$\begin{aligned} z &= e^{i\pi/18}, e^{13i\pi/18}, e^{-11i\pi/18} \\ \text{or } &e^{5i\pi/18}, e^{17i\pi/18}, e^{-7i\pi/18} \end{aligned}$$

An  $N$ 'th order polynomial has  $N$  roots.

Aside., what happens to  $z^{1/N}$  as  $N \rightarrow \infty$ ?

Interpret this as

$$\begin{aligned}(e^{\log z})^{1/N} &= e^{\frac{\log z}{N}} \\ &\approx 1 + \frac{\log z}{N} + \mathcal{O}(1/N)\end{aligned}$$

as  $N \rightarrow \infty$ , so  $N(z^{1/N} - 1) \sim \log z$ . This suggests we might be interested in logarithms of complex numbers...

## 5.1 Complex Logarithms

What is  $\log z$ ?

E.g.  $\log(2 + 4i)$ ? Write in polar form:

$$\begin{aligned}\log z &= \log(re^{i\theta}) \\ &= \log r + \log(e^{i\theta}) \\ &= \log r + \log(e^{i\theta} \cdot e^{2k\pi i}) \\ &= \log r + i\theta + 2k\pi i\end{aligned}$$

$\log z$  has infinitely many “answers”, it is a multi-valued function.

So  $\log(2 + 4i) = \log(re^{i\theta})$  where  $r = \sqrt{2^2 + 4^2} = \sqrt{20}$ .

$$\left. \begin{aligned}r \cos \theta &= 2 \implies \cos \theta = \frac{2}{\sqrt{20}} = 1/\sqrt{5} \\ r \sin \theta &= 4 \implies \sin \theta = \frac{4}{\sqrt{20}} = 2/\sqrt{5}\end{aligned} \right\} \text{ defines } \theta$$

So  $\log(2 + 4i) = \log(\sqrt{20}) + i[\theta + 2k\pi]$ .

Now we can solve many problems...

**Examples 5.2.**  $\sin x = 3$ . This has no solutions for  $x \in \mathbb{R}$ , but what about  $x \in \mathbb{C}$ ?

(a) “Express  $\sin x$  in exponential form”

(1)  $e^{ix} = \cos x + i \sin x$  (even if  $x$  is complex)

(2)

$$\begin{aligned}e^{-ix} &= \cos(-x) + i \sin(-x) \\ &= \cos x - i \sin x\end{aligned}$$

(1) and (2)  $\implies$

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix}) \equiv \cosh(ix)$$

$$\sin x = \frac{1}{2i}(e^{ix} - e^{-ix}) \equiv \frac{\sinh(ix)}{i}$$

$$\begin{aligned}
 \sin x = 2 &\iff \frac{1}{2i}(e^{ix} - e^{-ix}) = 2 \\
 &\implies u - \frac{1}{u} = 4i, \text{ where } u = e^{ix} \\
 &\implies u^2 - 4iu - 1 = 0 \\
 &\implies u = \frac{4i \pm \sqrt{(4i)^2 + 4}}{2} = \frac{4i \pm \sqrt{-12}}{2} \\
 &= \left(2 \pm \frac{\sqrt{12}}{2}\right)i = (2 \pm \sqrt{3})i = e^{ix}
 \end{aligned}$$

So

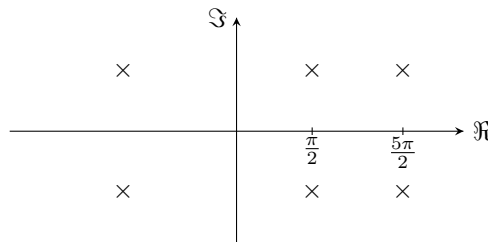
$$\begin{aligned}
 ix = \log u &= \log[(2 \pm \sqrt{3})i] \\
 &= \log(2 \pm \sqrt{3}) + \log i \\
 &= \log(2 \pm \sqrt{3}) + \frac{i\pi}{2} + 2k\pi i \\
 \implies x &= \frac{\pi}{2} + 2k\pi - i \log(2 \pm \sqrt{3})
 \end{aligned}$$

N.B. What can we say about  $\log(2 - \sqrt{3})$  and  $\log(2 + \sqrt{3})$ ?

$$\begin{aligned}
 \log(2 + \sqrt{3}) + \log(2 - \sqrt{3}) &= \log[(2 + \sqrt{3})(2 - \sqrt{3})] \\
 &= \log(4 - 3) = \log 1 = 0
 \end{aligned}$$

So we could write

$$x = \frac{\pi}{2} + 2k\pi \pm i \log(2 + \sqrt{3})$$



What is the radius of convergence of the power series for the function

$$f(x) = \frac{1}{2 - \sin x}?$$

$f(x)$  has singularities (poles / infinities) when  $\sin x = 2$ . Which means in the complex plane, the radius of convergence  $R$ , is

$$R^2 = \frac{\pi^2}{2} + (\log(2 + \sqrt{3}))^2$$

Return to  $\sin x = 2$ . How else might we solve this?

Lecture 27

(b) Alternatively, we could write  $x = u + iv$ , where  $u, v$  are real.

$$\sin(u + iv) = 2$$

i.e.

$$\begin{aligned}\sin u \cos(iv) + \cos u \sin(iv) &= 2 \\ \sin u \cosh v + i \cos u \sinh v &= 2\end{aligned}$$

$u, v$  are real.

Any complex equation is really two encoded real equations: the real parts and the imaginary part must balance.

$$\left. \begin{aligned}\sin u \cosh v &= 2 \\ \cos u \sinh v &= 0\end{aligned}\right\}$$

$\cos u \sinh v = 0 \implies$  either  $\cos u = 0$  or  $\sinh v = 0$ .

If  $\sinh v = 0 \implies v = 0$ , since  $\sinh$  is an increasing function  $\implies \cosh v = 1 \implies \sin x = 2$ . So no solutions for real  $x$ .

So we must have  $\cos u = 0 \implies u = \frac{\pi}{2} + n\pi$  for  $n \in \mathbb{Z} \implies \sin u = \pm 1$ , in particular  $\sin u = (-1)^n$ . The first equation says  $\sin u \cosh v = 2 \implies \cosh v = \pm 2 = 2(-1)^n$ , which has a solution only if  $n$  is even. So we write  $n = 2k$ . Then  $v = \pm \cosh^{-1} 2$ . Hence

$$x = \frac{\pi}{2} + 2k\pi + 2 \cosh^{-1} 2$$

Compare our earlier answer... It involved logs.

Recall

$$\begin{aligned}\cosh^{-1} v &= \log(v + \sqrt{v^2 - 1}) \\ \sinh^{-1} v &= \log(v + \sqrt{1 + v^2})\end{aligned}$$

So  $\cosh^{-1} 2 = \log(2 + \sqrt{3})$ , so  $x = \frac{\pi}{2} + 2k\pi \pm i \log(2 + \sqrt{3})$  - the same answer as before.

*Aside.* Complex numbers are fantastic as a means for solving real problems, E.g. To solve a cubic

$$x^3 + ax^2 + bx + c = 0$$

Write  $y = x + d$  so that  $a \rightarrow 0$ , then

$$y^3 + By + C = 0$$

Recall that

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$$

From de Moivre's Theorem from M1F:  $(e^{i\theta})^n = e^{in\theta}$ . So

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

Putting  $n = 3$  and taking the real part:

$$\begin{aligned}\cos(3\theta) &= \cos^3 \theta + 3i^2 \cos \theta (\sin \theta)^2 \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$

So  $\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta$ . Try writing  $y = M \cos \theta$ :

$$\begin{aligned}\implies M^3 \cos^3 \theta + BM \cos \theta + C &= 0 \\ \implies 4 \cos^3 \theta + \frac{4B}{M^2} \cos \theta + \frac{4C}{M^3} &= 0\end{aligned}$$

Choose  $M$  such that  $\frac{4B}{M^2} = -3$  - maybe  $M$  is imaginary. Then

$$\begin{aligned}4 \cos^3 \theta - 3 \cos \theta &= -\frac{4C}{M^3} \\ \implies \cos(3\theta) &= -\frac{4C}{M^3} \\ \implies \theta = \dots \implies y &= \dots\end{aligned}$$

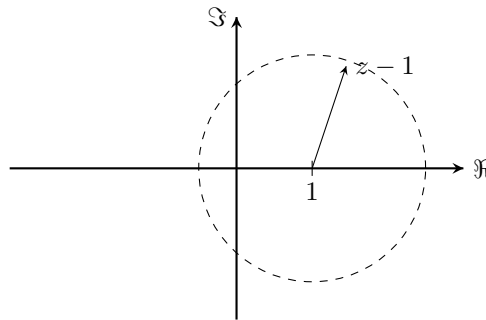
That is one way to obtain a not very useful formula for the solution to the general cubic - which is not worth knowing.

## 5.2 Plotting Curves in the Complex Plane

Any complex equation may have solutions which may define a curve. If so, we can plot it (maybe). Write  $z = x + iy$ .

### Examples 5.3.

(a)  $|z - 1| = 3$ . By geometry,  $z - 1$  has a fixed length so defines a circle of centre  $(1, 0)$  with radius 3.



Alternatively by algebra,  $z = x + iy$ , so

$$\begin{aligned}|z - 1| &= |x + iy - 1| \\ &= |(x - 1) + iy| \\ &= \sqrt{(x - 1)^2 + y^2} = 3\end{aligned}$$

$$\implies (x-1)^2 + y^2 = 3^3$$

$$(b) \left| \frac{z+2}{z-i} \right| = 2.$$

By algebra  $z = x + iy$ , so

$$\begin{aligned} |x+2+iy|^2 &= 4|x+iy-i|^2 \\ \implies x^2 + 4x + 4 + y^2 &= 4(x^2 + (y-1)^2) \\ \implies x^2 + 4x + 4 + y^2 &= 4x^2 + 4y^2 - 8y + 4 \\ \implies 0 &= 3x^2 - 4x + 3y^2 - 8y \\ \implies 0 &= (x - \frac{2}{3})^2 + (y - \frac{4}{3})^2 - \frac{20}{9} \end{aligned}$$

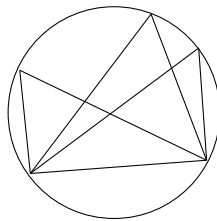
So we have a circle with centre  $(\frac{2}{3}, \frac{4}{3})$  and radius  $(\frac{2}{3}\sqrt{5})$ .

$$(c) \text{ What about } \text{Arg} \left( \frac{z+2}{z-i} \right) = \frac{\pi}{4}?$$

Geometrically, note that

$$\begin{aligned} \text{Arg} \left( \frac{z_1}{z_2} \right) &= \text{Arg} \left( \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} \right) \\ &= \text{Arg} \left( \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \right) \\ &= \theta_1 - \theta_2 \\ &= \text{Arg}(z_1) - \text{Arg}(z_2) \end{aligned}$$

So  $\text{Arg}(z+2) - \text{Arg}(z-i) = -\frac{\pi}{4}$ . So  $\text{Arg}(z-i) - \text{Arg}(z+2) = \frac{\pi}{4}$ , so the curve is the upper part of a circle:



## 6 First order Differential Equations

For the last couple lectures he just went over his handout on First order ODEs... this is covered in the first few pages of M1M2.

**- End of Mathematical Methods I -**